

Graph Prompt Learning: A Comprehensive Survey and Beyond

Abstract—Recent advances in foundation models have reshaped natural language processing and computer vision, but adaptation paradigms for graph-structured data remain less mature. Graph prompt learning has emerged as a promising approach for adapting pre-trained graph models with limited parameter updates, while also raising graph-specific questions about topology, task alignment, and transferability. This survey reviews graph prompt learning from the perspectives of cross-task, cross-domain, and cross-modal transfer. We first relate graph prompting to graph pre-training and then introduce a unified analytical framework covering prompt tokens, token structures, and insertion patterns. Based on this framework, we summarize representative methods for graph task prompting, multi-modal graph prompting, and graph domain adaptation. We further discuss application scenarios in online social networks, recommender systems, knowledge management, and biomedical graph learning, together with supporting benchmark and toolkit resources such as ProG and curated graph prompting repositories. Finally, we identify open challenges in graph foundation models, transfer protocols, prompt evaluation, theoretical understanding, and interpretability. Relevant resources are available at <https://github.com/WxxShirley/Awesome-Graph-Prompt> and <https://github.com/sheldonresearch/ProG>.

Index Terms—graph prompt learning, graph pre-training, graph representation learning, graph foundation models, transfer learning.

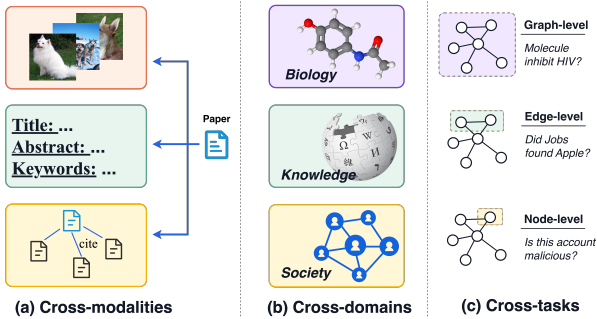


Fig. 1: Three recurring challenges for more general graph learning systems: (a) cross-modal transfer, (b) cross-domain transfer, and (c) cross-task transfer.

1 Introduction

Recent advances in foundation models have substantially reshaped research and applications in Natural Language Processing (NLP) and Computer Vision (CV). In contrast, the development of similarly mature adaptation paradigms for graph-structured data remains comparatively limited [10, 1, 65, 115, 150]. This gap is important because graphs are a natural abstraction for relational and structured data, and they play a central role in applications such as molecular discovery, drug design, recommender systems, and scientific modeling [63, 159, 25, 89, 84, 114]. As a result, there is growing interest in general-purpose graph learning systems that can better transfer across tasks, domains, and modalities.

However, achieving this goal remains difficult. Figure 1 highlights three recurring technical challenges for graph learning research. (1) How can models generalize across modalities? Recent multimodal foundation models have shown strong capabilities in aligning and reasoning across textual and visual inputs [10, 150, 1]. In graph-centered settings, however, com-

parable progress remains much more limited, especially when graph structure must be integrated with textual, visual, or scientific signals [59]. (2) How can models generalize across domains? Transfer learning has been effective in many machine learning settings, but transferring knowledge between graph domains remains challenging because both semantic spaces and structural patterns may shift substantially [170, 163]. This makes graph domain adaptation a difficult and still open research problem. (3) How can models generalize across tasks? A large portion of the existing literature focuses on transferring pre-trained graph knowledge across tasks within the same graph domain, such as node classification, link prediction, and graph classification [101, 68, 103, 13, 169, 40, 96, 21]. Yet, compared with NLP and CV, standardized real-world evaluations and deployment-oriented studies for graph prompting remain relatively limited. Taken together, these observations suggest that graph prompting should be studied not only as a task adaptation trick, but also as part of a broader effort toward more general and reusable graph learning systems.

These challenges have made parameter-efficient adaptation strategies increasingly attractive [19, 56]. Among them, prompt learning has emerged as a practical mechanism for adapting pre-trained models without repeatedly updating all model parameters [85, 109, 65]. Prompting modifies the input or task formulation so that downstream problems better align with behaviors induced during pre-training. In many settings, this enables practitioners to reuse pre-trained models with substantially fewer trainable parameters than full fine-tuning [1, 46, 56, 95]. For graph learning, this property makes prompting a promising, though not universally sufficient, mechanism for task adaptation, cross-domain transfer, and multimodal interaction.

From a representational perspective, several data modalities can be reinterpreted through graph-structured abstractions. A sentence may be modeled as a dependency graph or token interaction graph, and an image may be represented as

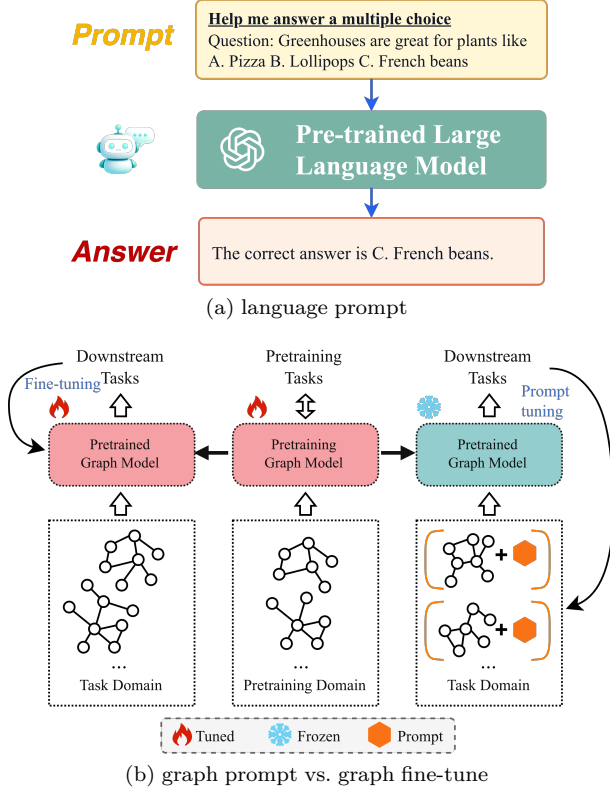


Fig. 2: Language prompt vs. graph prompt. A textual prompt modifies the input so that a pre-trained language model can perform a new task, such as multiple-choice question answering. A graph prompt follows the same adaptation intuition, but it must operate on graph features, structures, or task interfaces rather than on text alone.

a grid graph or region graph. This perspective motivates the adaptation of successful prompting ideas from language and vision to graph learning [101, 68, 103, 13, 169, 74, 24, 3, 22].

Although graph prompting is partly inspired by prompting techniques in NLP, the two are not directly analogous [103]. First, the design space of graph prompts is substantially richer. In language models, prompts may take the form of discrete templates, soft prompt vectors, prefixes, or instruction-style conditioning [1, 19]. In graph learning, however, a prompt often also involves structural design choices, insertion mechanisms, and answering functions, rather than prompt content alone. Second, aligning downstream graph problems with pre-training objectives is typically more difficult than in language tasks [68, 103]. A typical language pre-training task such as masked word prediction [10] shares a relatively direct interface with many downstream tasks [65], whereas graph tasks span node-level [23], edge-level [153], and graph-level objectives [99, 102], making pre-training pretexts less directly reusable. Third, graph prompts are usually less interpretable to non-specialists than natural-language prompts, because their effects are mediated by graph topology, feature transformations, and message passing. These differences make it necessary to examine graph prompting on its own terms rather than treating it as a straightforward extension of NLP prompting.

The literature on graph prompting has expanded rapidly in recent years, but it remains fragmented across task settings, graph modalities, and adaptation strategies. Existing

works often emphasize different perspectives, making direct comparison difficult and slowing the systematic accumulation of knowledge in this area. This survey therefore provides a structured review of graph prompting through the lens of the three transfer challenges discussed above, while also clarifying its relationship to graph pre-training, graph foundation models, and multimodal graph learning. To organize the field, we consider the following research questions (RQs):

- RQ1: How can existing graph prompting methods be organized under a unified analytical framework? We seek a common vocabulary for comparing methods that differ in prompt representation, insertion mechanism, and adaptation objective.
- RQ2: What functional role do prompts play in graph representation learning? We study what graph prompts actually contribute, how they interact with graph learners, and why they may behave differently from conventional fine-tuning.
- RQ3: What are the main design dimensions of graph prompts? We examine how graph prompts are constructed, how they align downstream tasks with pre-training objectives, and how they are optimized in practice.
- RQ4: How can graph prompting be supported in practical applications and software ecosystems? We discuss the current state of toolkits, benchmarks, and reusable implementations for graph prompting in real-world scenarios.
- RQ5: What limitations remain in current graph prompting research, and which future directions appear most promising? We summarize open challenges and outline research directions that may shape the next stage of the field.

RQ1 motivates a unified framework for analyzing graph prompt learning. This framework decomposes graph prompts into prompt tokens, token structures, and inserting patterns, thereby providing a stable lens for comparing existing variants.

RQ2 concerns the relationship between prompts and existing graph models through the lenses of flexibility and expressiveness, leading to a deeper understanding of the role of graph prompts. Unlike many prompt learning surveys in NLP [65], which treat prompting primarily as a mechanism for bridging pre-training and downstream tasks, we discuss graph prompting as being intertwined with the modeling capacity of graph learners themselves. This perspective helps clarify why prompting is promising in the graph domain and how it differs from conventional fine-tuning methods [39].

RQ3 motivates a taxonomy that organizes representative graph prompting studies according to node-level, edge-level, and graph-level settings, while aligning them with the broader context of pre-training objectives. This organization clarifies prompt mechanisms within the overall “pre-training and prompting” workflow.

RQ4 concerns the supporting ecosystem for graph prompting. We discuss benchmark and toolkit resources such as ProG¹ and curated repositories² that improve accessibility, repro-

1. <https://github.com/sheldonresearch/ProG>

2. <https://github.com/WxxShirley/Awesome-Graph-Prompt>

ducibility, and comparability for researchers and practitioners working on graph prompting.

Beyond these questions, the survey further discusses application scenarios, current challenges, and future research directions, thereby outlining a broader roadmap for the development of graph prompt learning (RQ5). The contributions are summarised as follows:

- **Enabling Comprehensive Analysis.** We develop a unified framework for analyzing graph prompt learning, providing a structured view of prompt tokens, token structures, and inserting patterns.
- **Prompt-Model Perspective.** We discuss graph prompts not only as task adapters but also as mechanisms that interact with the flexibility and expressiveness of graph learners.
- **A Systematic Taxonomy of Graph Prompting.** We synthesize representative graph prompting studies and organize them within the overarching “pre-training and prompting” workflow.
- **Empowering the Graph Prompting Ecosystem.** We summarize toolkits and community resources that support reproducible graph prompting research.
- **Charting a Path Forward.** We provide a detailed discussion of current challenges and future directions in the field.

Roadmap. The remainder of this survey is organized as follows: Section 2 presents the survey methodology, followed by preliminaries in Section 3, pre-training methods in Section 4, and prompting methods for graph models in Section 5. Section 6 then discusses graph prompting in multi-modal and multi-domain settings. Section 7 discusses potential applications of graph prompting, and Section 8 presents ProG. Section 9 discusses current challenges and future directions. Section 10 concludes the survey.

2 Survey Methodology

2.1 Research Objectives

This survey studies graph prompting as a route toward more transferable and reusable graph learning systems. We organize the discussion around the three transfer challenges summarized in Table 1: transfer across modalities, domains, and tasks. Prompting has become an important adaptation mechanism in NLP and vision [1, 19, 115], but its role in graph learning is still less settled. Our goal is therefore not to assume that prompting is a universal solution, but to examine when and how graph prompts help graph models transfer across tasks and domains, and how they interact with other modalities such as text and images. Beyond these shared transfer challenges, graph prompting also differs from its counterparts in NLP and CV, which motivates the more specific research questions listed in Table 1.

2.2 Taxonomy

Figure 3 summarizes how this survey organizes the graph prompting literature. The taxonomy is designed to show how prompt-based methods relate to pre-training objectives, transfer settings, and application scenarios in graph learning.

- (1) Pre-training Strategies for Graph Prompt.

Prompting methods are often motivated by the need to align downstream tasks with behaviors induced during pre-training. For this reason, the survey first reviews representative pre-training strategies in graph learning before turning to prompting itself. We group existing pre-training approaches into node-level, edge-level, graph-level, and multi-task settings, and then examine how later prompting methods reuse or adapt these objectives. This setup makes it easier to compare prompting methods against the pre-training assumptions on which they depend.

(2) Prompting Methods in Graph Learning.

To cover the three transfer challenges defined in Table 1, we analyze graph prompting from three directions: i. prompt design for graph tasks (Section 5); ii. multi-modal prompting with graphs (Section 6); and iii. graph domain adaptation with prompting techniques (Section 6.2). Across these directions, the detailed questions in Table 1 connect method design, adaptation mechanism, practical scope, and open challenges.

i. **Prompt Design for Graph Tasks.** In this part, we introduce a unified analytical framework for graph prompt design. We organize existing methods around three components: prompt tokens, which represent prompt content; token structures, which specify how multiple prompt units are organized; and inserting patterns, which determine how prompts are combined with the original graph. We then discuss answering functions and optimization strategies as follow-up issues that affect task alignment and adaptation. This organization lets us compare methods in terms of shared assumptions, design differences, and trade-offs.

ii. **Multi-modal Prompting with Graphs.** In this part, we discuss prompting methods for text-attributed and other multi-modal graph settings. The recent growth of graph-LLM and graph-language research has expanded this area substantially, so our goal here is not to provide a full survey of graph-LLM methods. Instead, we focus on prompting-oriented interfaces and adaptation mechanisms that explicitly involve graph structure.

iii. **Graph Domain Adaptation with Prompting.** We then examine how prompts are used when the pre-training and downstream graphs come from different domains. This part separates semantic alignment from structural alignment, because feature meanings and graph topology often shift in different ways. This organization also clarifies why cross-domain graph prompting cannot be reduced to ordinary task adaptation.

2.3 Literature Overview

This survey covers representative graph prompting papers published through early 2026, with emphasis on conference and journal articles as well as influential arXiv preprints. The surveyed literature includes work from venues such as NeurIPS, SIGKDD, The Web Conference, ICLR, CIKM, ICML, IJCAI, EMNLP, SIGIR, ACL, AAAI, WSDM, and TKDE. We also include recent preprints when they had visible influence on subsequent literature, benchmarks, or public implementations before formal publication. Figure 4a summarizes the venue distribution of the collected papers, and Figure 4b shows the most frequent keywords in their titles.

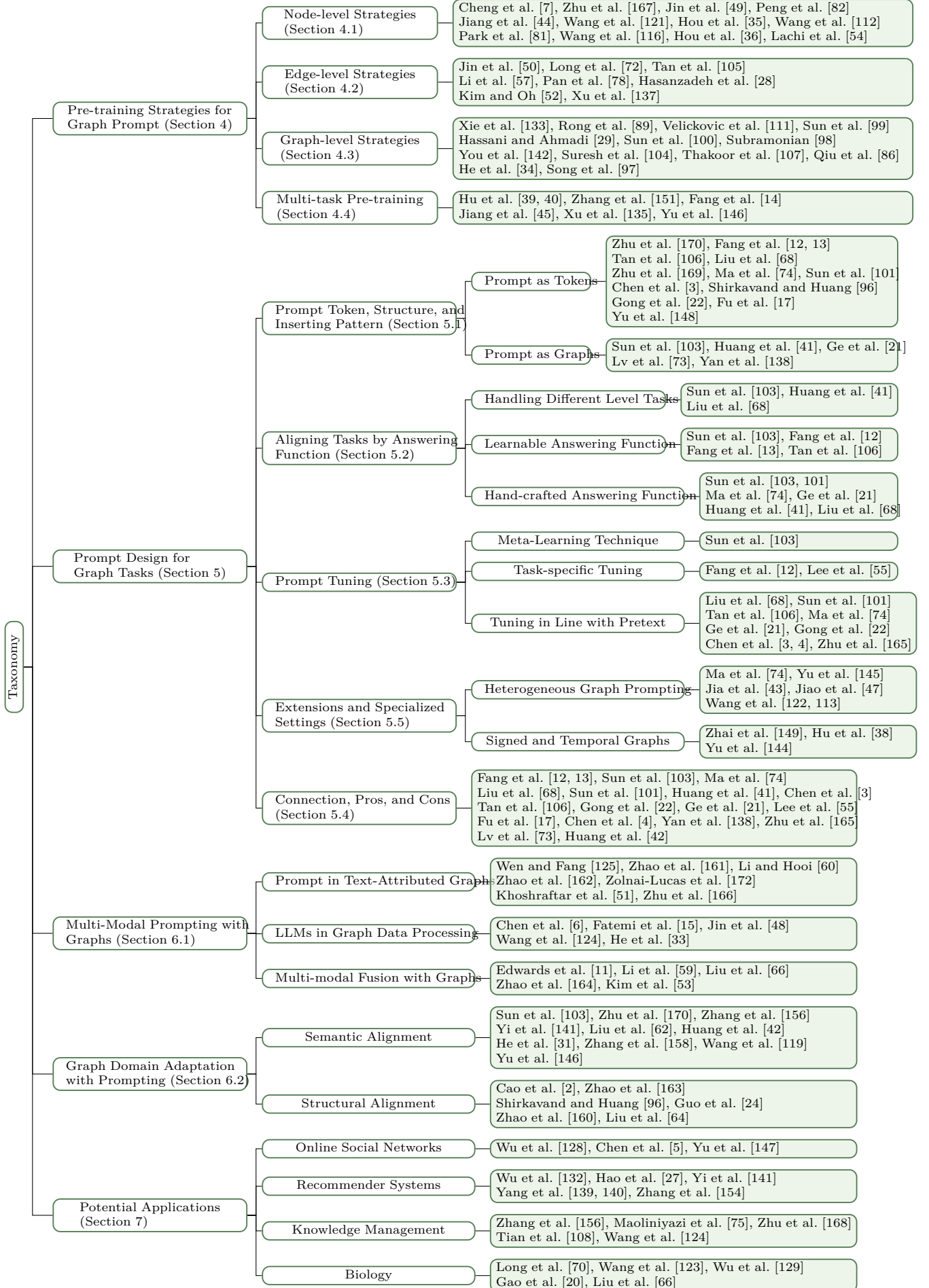


Fig. 3: Taxonomy of this Survey with Representative Works.

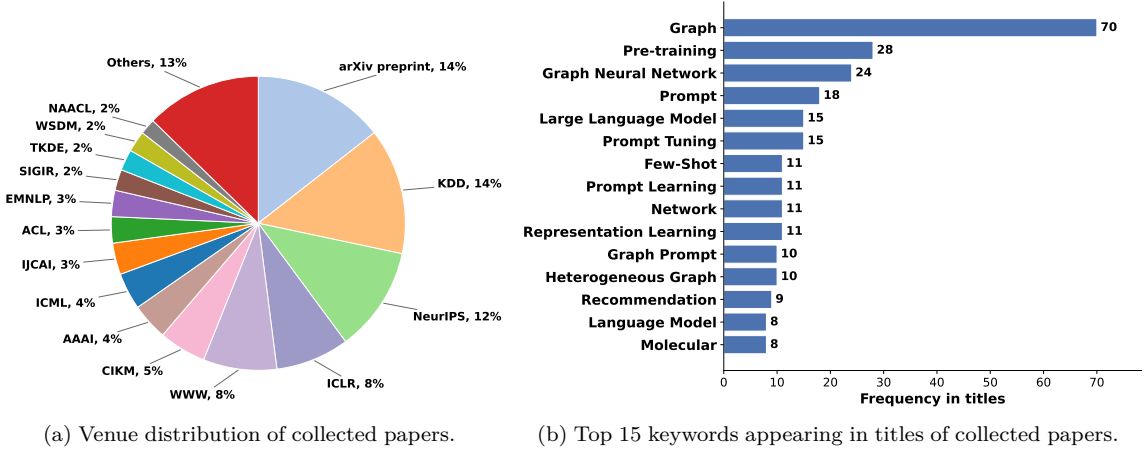


Fig. 4: Statistics of the collected papers

TABLE 1: Research Questions and Scope

Transfer Challenges in Graph Learning	P1: How to Make the Model General for Different Modalities? (Section 6.1)
	P2: How to Make the Model General for Different Domains? (Section 6.2)
	P3: How to Make the Model General for Different Tasks? (Section 5)
Detailed Questions of Graph Prompt	Q1: How to Understand Existing Work with a Unified Framework? (Section 5.1)
	Q2: What’s the Nature of Graph Prompt? (Section 3.5, Section 5.4)
	Q3: How to Design Graph Prompts? (Section 5)
	Q4: How to Deploy Graph Prompts in Real-world Applications? (Section 7, Section 8)
	Q5: What Are the Current Challenges and Future Directions? (Section 9.1)

2.4 Connection to Existing Work

Recent surveys on graph prompt learning differ mainly in scope and organizing principle. Wu et al. [131] reviewed the early wave of graph prompting methods with an emphasis on techniques, applications, and open challenges when the methodological space was still relatively compact. Long et al. [71] expanded the discussion to design principles, application scenarios, benchmark issues, and prompt tuning strategies. Fu et al. [18] organized the literature around the “pre-train and prompt” workflow and emphasized data-level, representation-level, and task-level perspectives.

Our emphasis differs in two respects. First, we adopt a unified decomposition of graph prompting into prompt tokens, token structures, and insertion patterns, while further discussing answering functions and optimization strategies as task-alignment and adaptation issues. Second, we place graph prompting within a broader taxonomy that also includes pre-training strategies, multi-modal prompting, and graph domain adaptation. We further retain a practical perspective through the discussion of application scenarios and supporting benchmark/toolkit resources such as ProG. Related surveys on graph foundation models, graph-LLM integration, and graph self-supervised learning [63, 61, 134] provide broader background, whereas this survey concentrates specifically

on graph prompting and its surrounding adaptation mechanisms.

3 Preliminaries

This section introduces the background needed for the rest of the survey. We first define the basic notation used throughout the paper, then review the main stages of graph representation learning, including shallow embeddings, supervised GNNs, and pre-training-based transfer. We also briefly revisit the development of prompt learning and explain why prompting deserves separate discussion in graph learning rather than being treated as a direct copy of NLP practice.

3.1 Notations

Let a graph be denoted by $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ is the node set and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set. Each node v_i is associated with a feature vector $\mathbf{x}_i \in \mathbb{R}^D$. We use $\mathbf{X} \in \mathbb{R}^{N \times D}$ to denote the node feature matrix and $\mathbf{A} \in \{0, 1\}^{N \times N}$ to denote the adjacency matrix, where $\mathbf{A}_{ij} = 1$ if and only if $(v_i, v_j) \in \mathcal{E}$.

3.2 Graph Representation Learning

Graph representation learning has developed through several overlapping stages. Early work focused on shallow embedding methods, which map nodes to learnable low-dimensional vectors and are then reused in downstream tasks. Representative examples include DeepWalk [83] and node2vec [23]. These methods are flexible because the learned embeddings can often be reused across multiple tasks, but they are usually transductive and do not naturally incorporate rich node attributes.

Shallow embedding methods are often grouped by how they define node similarity. One common line is factorization-based [77, 152]; another is based on random walks [23, 83, 120, 118]. These methods remain useful when the goal is to preserve structural proximity in a compact representation, but their transfer behavior is limited once node attributes, unseen nodes, or task-specific supervision become central.

Graph neural networks (GNNs) address some of these limitations by learning task-relevant representations through parameterized neighborhood aggregation. Most GNNs follow a

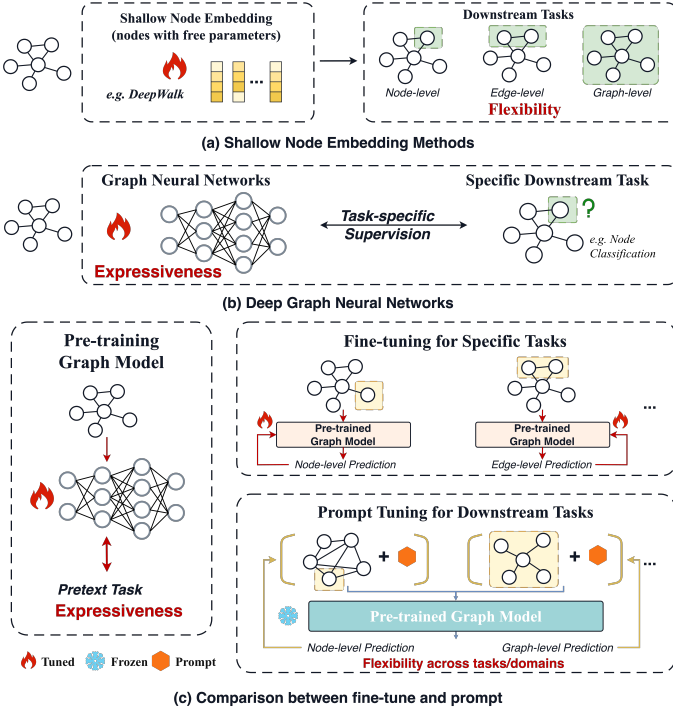


Fig. 5: A schematic view of flexibility and expressiveness in graph learning. (a) Shallow node embedding methods are often reusable across tasks but have limited modeling capacity. (b) Task-specific GNNs are more expressive but usually require re-training or supervision for new tasks. (c) Prompting can improve the trade-off by adapting the input while reusing a pre-trained graph model.

message-passing framework and use a more expressive encoder than shallow embeddings. Representative architectures include spectral and spatial variants [9, 32, 76, 26, 110, 94, 136]. In many settings, GNNs outperform shallow embeddings because they can combine graph structure with node features. The trade-off is that they are usually trained for specific tasks and often require supervision or repeated re-training when the task changes.

This contrast motivates the transfer problem that runs through the rest of the survey. Shallow embeddings are flexible but limited in modeling power. Task-specific GNNs are more expressive but less reusable across settings. Pre-training and prompting can both be understood as attempts to improve transfer, albeit through different mechanisms.

3.3 Pre-training and Fine-tuning

The pre-training and fine-tuning workflow has become a common transfer paradigm in graph representation learning. The usual workflow is to pre-train a model on large-scale graph data, with or without labels, and then adapt it to downstream tasks by updating model parameters. In graph learning, representative pre-training objectives include graph autoencoding [112], masked component modeling [39, 89], and graph contrastive learning [111, 99]. We review these objectives in detail in Section 4.

This paradigm improves transfer because the downstream model no longer starts from scratch. Still, fine-tuning remains model-centric: the usual way to adapt to a new task is

to update some or all of the model parameters. This can work well, but it also means that transfer often depends on repeated optimization, access to supervision, and a stable match between the new task and the pre-trained model.

3.4 A Brief History of Prompt Learning

Prompt learning emerged as another way to adapt pre-trained models, especially when full fine-tuning became increasingly expensive. Rather than updating the whole model for every downstream problem, prompting changes the input, the task interface, or a small set of learnable prompt parameters so that the downstream problem is presented in a form closer to what the model has already seen during pre-training [65].

In NLP, prompts take several forms. Some are discrete templates such as cloze-style prompts. Others are learned continuous prompts, including soft prompt tuning [56] and prefix tuning [58]. Prompt construction may rely on manual templates [1, 91, 92], discrete prompt search [46, 30, 95, 19], or prompting directly in the embedding space [58, 56, 109, 85]. These developments do not replace fine-tuning altogether. Instead, they expand the set of available adaptation mechanisms.

For graph learning, prompting is attractive for a similar reason: it may enable transfer with fewer updated parameters and with a clearer separation between the pre-trained graph encoder and the downstream adaptation module. At the same time, graph prompting is not a direct replica of NLP prompting, because graph inputs are structured objects and prompt design must also account for topology, insertion patterns, and graph-level prediction mechanisms.

3.5 Why Graph Prompt?

A. Differences between Graph Prompting and Graph Fine-tuning. The difference between graph prompting and graph fine-tuning is easiest to state in terms of where adaptation happens. Fine-tuning is primarily a model-level adaptation strategy: a pre-trained graph model is adjusted so that its parameters better fit a downstream task. Prompting, by contrast, is primarily an input-level or data-level adaptation strategy: the downstream graph is modified through prompt tokens, prompt structures, or prompt-induced interfaces before it is processed by a pre-trained model. In practice, graph prompting may still involve a small number of trainable parameters, but the central idea is not to re-optimize the full encoder for each new task.

B. Prompts as Learnable Graph Modifications. One useful way to understand graph prompting is to view it as a learnable approximation to graph transformation. Let φ^* denote a pre-trained graph model, $\mathcal{G}$ the original graph, $\mathcal{G}_p^*$ a prompt graph or prompt module, and $\psi(\mathcal{G}, \mathcal{G}_p^*)$ the prompted graph produced by inserting the prompt into the input. Let $\mathbf{g}(\cdot)$ denote a target graph transformation, so that $\mathbf{g}(\mathbf{A}, \mathbf{X})$, acting on the adjacency matrix $\mathbf{A}$ and node feature matrix $\mathbf{X}$ of $\mathcal{G}$, yields a transformed graph of the same type as $\mathcal{G}$. Both $\psi(\mathcal{G}, \mathcal{G}_p^*)$ and $\mathbf{g}(\mathbf{A}, \mathbf{X})$ are thus valid graph inputs to φ^* . Then, at a conceptual level, graph prompting can be written as

$$\varphi^*(\psi(\mathcal{G}, \mathcal{G}_p^*)) = \varphi^*(\mathbf{g}(\mathbf{A}, \mathbf{X})) + O_{p\varphi}^*, \quad (1)$$

where $O_{p\varphi}^*$ is the residual discrepancy between the prompted input and the ideal transformation under the pre-trained

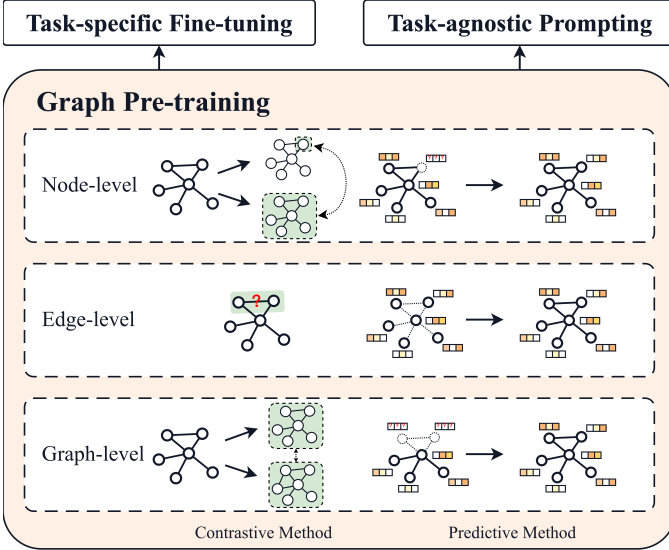


Fig. 6: Graph pre-training methods.

model φ^* . This expression should not be read as a formal guarantee. It is a conceptual lens: prompts can be interpreted as a learnable way to adjust graph structure, node features, or task interfaces so that the downstream input better matches the behavior already captured by the pre-trained encoder.

C. *Uses Beyond Task Adaptation*. This perspective also explains why graph prompting may be useful beyond ordinary task transfer. If a prompt can act as a learnable graph modification, then it may be used to adjust the input before inference, for example by compensating for structural mismatch, introducing auxiliary prompt nodes, or aligning a graph with another modality. In principle, this opens the door to applications beyond standard downstream adaptation, including domain alignment, multi-modal interfaces, and robustness-oriented input adjustment. Whether these uses help in practice remains task-dependent, but the key point is that prompting changes the way adaptation is expressed: not only through model updates, but also through learnable modifications to the input graph itself.

In short, graph prompting is worth separate study because it changes the unit of adaptation. Instead of relying solely on repeated model re-training, it offers a way to reuse a pre-trained graph encoder while expressing downstream adaptation through prompt parameters, prompt structure, and prompted graph construction. That shift is exactly what makes graph prompting both flexible and technically distinct from conventional fine-tuning.

4 Pre-training GNNs for Graph Prompting

Graph pre-training is a pivotal step of the pre-training, prompting, and predicting paradigm in graph representation learning. This approach leverages readily available information to encode the inherent graph structure, providing a robust foundation for generalization across diverse downstream tasks. By integrating these pre-training methods into the comprehensive workflow, we offer an exploration of their interplay with the subsequent prompting and predicting phases, shedding light on the strengths and limitations of this holistic approach. This unique perspective distinguishes

our survey, framing graph pre-training as an integral part of the broader graph-prompting learning process. To better illustrate the motivation behind the prompting paradigm, we will now delve into four distinct pre-training strategies within the traditional pre-training and fine-tuning framework.

4.1 Node-level Strategies

Node-level pre-training methods aim to learn transferable local representations for nodes and their neighborhoods. As shown in Figure 6, these methods are usually built around either contrastive or predictive objectives. In contrastive settings, self-supervised signals are typically created by perturbing graph structure or node attributes and then maximizing agreement between the original and augmented views. Representative node-level contrastive methods include [7, 167, 49, 82, 44, 121].

Predictive node-level methods instead reconstruct or corrupt information from the unperturbed graph. Representative examples include masked autoencoding and related reconstruction objectives [35, 112, 81, 116, 36]. These methods can learn useful local patterns, but their transferability still depends on how well the pretext objective matches the downstream task. In particular, methods centered on local corruption or reconstruction may under-emphasize long-range dependencies or graph-level semantics. Recent work has also extended node-oriented pre-training to multi-graph settings, which is more consistent with the graph foundation model view of transfer [54].

4.2 Edge-level Strategies

Edge-level pre-training focuses on node interactions and pairwise relations, which makes it particularly relevant to tasks such as link prediction. One common approach is to discriminate whether an edge should exist between a pair of nodes, which can be viewed as a contrastive or relational discrimination objective [50, 72]. Another approach reconstructs masked or corrupted edges, often through adjacency recovery or related generative objectives [105, 57, 78, 28, 52].

These strategies are useful when downstream performance depends strongly on local connectivity patterns. Their limitation is that they are usually centered on structure alone. As a result, they may not transfer well when node attributes, higher-order semantics, or graph-level objectives matter more than pairwise edge recovery. Recent benchmark work likewise evaluates transfer more broadly than edge reconstruction alone [137].

4.3 Graph-level Strategies

Graph-level pre-training aims to improve whole-graph representations, especially for downstream tasks defined on subgraphs or complete graph instances. Similar to node- and edge-level settings, graph-level strategies are often divided into reconstruction-based and contrastive approaches. Reconstruction-based methods mask graph components and learn to recover them [133, 89]. Contrastive methods instead maximize mutual information between local and global views [111, 99, 29, 100, 98], or align augmented graph views, with or without explicit negative pairs [142, 104, 107, 86].

These methods often produce useful graph-level representations, especially when the downstream objective is close

to the pretext task. The main difficulty is transfer across tasks and datasets with larger semantic gaps. When the pre-training signal is too narrowly aligned with one kind of graph structure, the resulting encoder may transfer poorly and can even suffer from negative transfer [90]. More recent work has pushed graph-level pre-training toward cross-domain and text-attributed graph settings, which broadens the scope of transfer beyond a single graph family [34, 97].

4.4 Multi-task Pre-training

Multi-task pre-training combines several objectives so that the encoder learns from multiple views of graph data at once. Typical choices include joint optimization of attribute reconstruction and structural recovery [39, 40, 151, 14]. For example, Hu et al. [39] pre-trained a GNN at both the node and graph levels so that local and global information could be learned together. Other work combines contrastive learning across multiple granularities [45, 135].

The advantage of multi-task pre-training is breadth: the encoder is less tied to a single pretext task. The drawback is optimization complexity. Different objectives may compete with each other, and the resulting model may underperform on some tasks even if aggregate pre-training loss improves. For this reason, multi-task pre-training is best understood as a broader design space rather than a guaranteed remedy for negative transfer. This line of thinking now overlaps with multi-graph and multi-domain pre-training, where the goal is to build encoders that transfer across graph collections rather than within a single benchmark only [54, 146].

4.5 Further Discussion

The above strategies show that graph pre-training can provide reusable encoders, but they also clarify why pre-training alone is not sufficient for broad transfer. A pre-trained model is useful only when its learned representations and pretext objectives remain compatible with downstream tasks, domains, and graph structures. Prompting offers one way to reduce this mismatch: instead of fully re-optimizing the encoder for every new setting, a graph prompt can adjust the input graph, task interface, or readout process so that the downstream problem is closer to the behavior induced during pre-training. This does not make prompting a universal replacement for fine-tuning, since its effectiveness still depends on prompt design, pre-training diversity, and the size of the distribution shift. It does, however, provide a practical adaptation mechanism for settings where supervision is limited or repeated full fine-tuning is expensive.

5 Prompt Design for Graph Tasks

After reviewing graph pre-training objectives, we now turn to how prompts are designed and used to adapt pre-trained graph models. As shown in Figure 7, a graph prompt usually contains three components: prompt tokens with prompt vectors, token structures that preserve correlations among these tokens, and inserting patterns that specify how prompts are integrated with the original graph. Based on these components, and following Q1 and Q3 in Section 2, we analyze graph prompt design along four aspects: how a graph prompt is constructed; how downstream tasks are reformulated toward

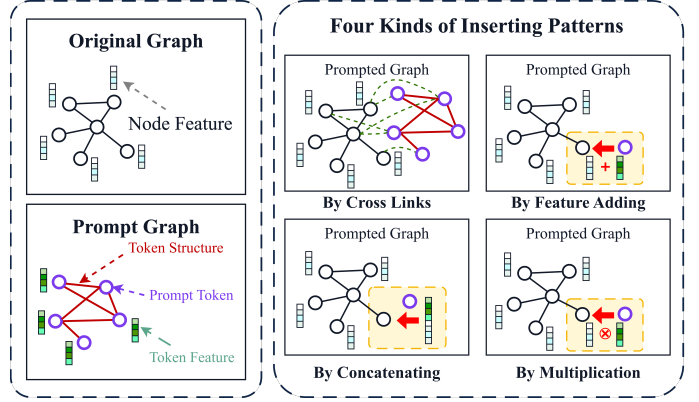


Fig. 7: Prompt Tokens, Structures, and Inserting Patterns.

the pretext; how a graph prompt is learned; and what trade-offs connect existing designs. With these aspects, we summarize representative methods in Table 2.

5.1 Prompt Token, Structure, and Inserting Pattern

How do they design a graph prompt?

A. Prompt as Tokens. The simplest graph prompts can be represented as additional feature vectors attached to the original graph features [170]. Given a graph feature matrix $\mathbf{X} = \{x_1, \dots, x_N\} \in \mathbb{R}^{N \times d}$, where $x_i \in \mathbb{R}^{1 \times d}$ denotes the feature of node v_i , Fang et al. [12] and Shirkavand and Huang [96] define a learnable prompt vector $p \in \mathbb{R}^{1 \times d}$ and add it to each node feature, producing $\mathbf{X}^* = \{x_1 + p, \dots, x_N + p\}$. The prompted feature matrix is then fed into a pre-trained graph model. GPF-Plus [13] extends this idea from a single shared token to multiple basis tokens, which increases the capacity of feature-level prompting while keeping the prompt lightweight.

Other methods use token prompts in more specialized ways. VNT [106] concatenates prompt tokens with the original node set and lets a graph transformer integrate them through self-attention, so the prompt interaction is learned inside the transformer rather than through explicit prompt-node edges. GraphPrompt [68] places the prompt in the hidden representation space and uses it to guide the readout operation. Given node embeddings $\{h_v : v \in \mathcal{V}\}$ and a task-specific prompt token p_t , it computes $s_t = \text{Readout}(\{p_t \otimes h_v : v \in \mathcal{V}\})$. SGL-PT [169] introduces a prompt token connected to all graph nodes, so that the prompt can provide global information for graph classification. GPPT [101] defines task tokens and structure tokens to align node classification with link prediction: the label is represented as a task token, the target node neighborhood is represented as a structure token, and prediction is reformulated as estimating the link between the two.

Later token-based designs further relax the assumption that prompts must be a single vector shared by all nodes. ULTRA-DP [3] creates a target-node-specific prompt by combining positional embeddings with task embeddings from pre-training. HetGPT [74] uses node tokens and class tokens, and adds type-specific feature tokens to handle heterogeneous graphs. Self-Pro [22] constructs semantic and structural prompts from the input graph and reuses the pre-trained projector as a downstream adapter. EdgePrompt [17] shifts part of the prompt from nodes to edges and injects edge

TABLE 2: Summary of selected representative works on graph prompt. The pre-training columns indicate the task level explicitly used or assumed by each method’s upstream objective, and the downstream columns indicate task levels explicitly evaluated or supported in the original paper. Preset denotes a hand-crafted or rule-based answering function without a trainable task head, while Learnable denotes a learned classifier, decoder, or task head. $\mathcal{S}$: Subgraph. $V(\mathcal{S})$: Node set within subgraph $\mathcal{S}$. π : Pre-trained parameters. ϕ : Task head parameters. θ : Prompt parameters. Tilded variables and X^* denote prompted representations or prompted features.

	pre-training task			prompt design			downstream tasks			answering function	
Paper	node	edge	graph	prompt components	inserting pattern	prompt tuning	node	edge	graph	Preset	Learnable
GPPT (KDD 2022 [101])	✗	✓	✗	structure token: $s_v \in \mathbb{R}^d$ task token: $c_y \in \mathbb{R}^d$	$s_{v_i} \leftarrow f_\theta(v_i)$ $\tilde{s}_{y,v_i} \leftarrow [c_y, s_{v_i}]$	Cross Entropy	✓	✗	✗	✓	✗
GPF (arXiv [12])	✓	✓	✓	prompt feature $p \in \mathbb{R}^d$	$\tilde{s}_i \leftarrow x_i + p$	$\max_{p,\phi} \phi \sum (y_i, \tilde{s}_i)$ $p_{\pi,\phi}(y_i \tilde{s}_i)$	✗	✗	✓	✗	✓
All in One (KDD 2023 [103])	✗	✗	✓	prompt token: $\mathcal{P} = \{p_1, \dots, p_{ \mathcal{P} }\}$ token structure: $\{(p_i, p_j) p_i, p_j \in \mathcal{P}\}$	$w_{ik} \leftarrow \sigma(p_k \cdot x_i^T)$ if $> \delta$ $G' \leftarrow (V \cup \mathcal{P}, E \cup E_{\mathcal{P}} \cup E_{\text{cross}})$ $E_{\text{cross}} = \{(v_i, p_k, w_{ik})\}$	Meta-Learning	✓	✓	✓	✓	✓
GraphPrompt (WWW 2023[68])	✗	✓	✗	prompt token: $p_t \in \mathbb{R}^d, t \in \mathcal{T}$ structure token: $s \in \mathbb{R}^d$ task token: $c_y \in \mathbb{R}^d$	$\tilde{s}_i^t \leftarrow \text{Readout}(\{p_t \odot f_\pi(v) $ $v \in V(S_i)\})$ $c_y \leftarrow \text{Mean}(\{\tilde{s}_i^t y_i = y\})$	$\min_{p_t} - \sum (y_i, \tilde{s}_i) \ln$ $\frac{\exp(\text{sim}(\tilde{s}_i^t, c_y) / \tau)}{\sum_{y \in \mathcal{Y}} \exp(\text{sim}(\tilde{s}_i^t, c_y) / \tau)}$	✓	✗	✓	✓	✗
PRODIGY (NeurIPS 2023 [41])	✗	✓	✗	data graph: $\mathcal{G}^D \sim \oplus_{i=1}^K \mathcal{N}(V_i, G_i)$ task graph: $\mathcal{G}^T$	$\tilde{s}_i \leftarrow f_{\pi}^T(\mathcal{G}^T f_{\pi}^D(\mathcal{G}^D))$	Fixed	✓	✓	✓	✓	✗
GraphPromter (ICDE 2025 [73])	✗	✓	✗	candidate prompts: $\mathcal{G}^S = \{G_p\}_{p=1}^N$ task graph: $\mathcal{G}^T$ cache: $\mathcal{C}$	$\hat{S}' \leftarrow \hat{S} \cup \mathcal{C}$ $H \leftarrow \text{GNN}^T(G^T(\hat{S}', Q))$	$L_{NM} = \sum_{q \in Q_{NM}} \mathcal{L}_{CE}(P_q^{NM}, y_q^{NM})$ $L_{MT} = \sum_{q \in Q_{MT}} \mathcal{L}_{CE}(P_q^{MT}, y_q^{MT})$	✓	✓	✗	✓	✗
SGL-PT (arXiv [169])	✓	✗	✓	prompt token: one vector for each graph	connect to all nodes in the graph	contrastive loss and reconstruction loss	✗	✗	✓	✓	✗
GPF-Plus (NeurIPS 2023 [13])	✓	✓	✓	prompt features $p_1, \dots, p_k \in \mathbb{R}^d$	$\tilde{s}_i \leftarrow x_i + \sigma(p_1,$ $\dots, p_k)$	$\max_{p,\phi} \phi \sum (y_i, \tilde{s}_i)$ $p_{\pi,\phi}(y_i \tilde{s}_i)$	✗	✗	✓	✗	✓
RELIEF (KDD 2025 [165])	✓	✓	✓	feature prompt matrix: $P_t \in \mathbb{R}^{n \times d}$	$X^* \leftarrow X + P_t$ $P_t^{a,:} = (a_t, z_t)$	$r_t \leftarrow \mathcal{L}_{t-1} - \mathcal{L}_t$ H-PPO	✓	✗	✓	✗	✓
DeepGPT (arXiv [96])	✗	✗	✗	prompt token: $p \in \mathbb{R}^d$ prefix token: $P \in \mathbb{R}^{ \mathcal{P} \times d}$	$\tilde{x}_i \leftarrow x_i + p$ $\tilde{s}_i \leftarrow f_{P,\pi}(G, \tilde{x}_i)$	$\min_{p,P,\phi} \phi \sum (y_i, v_i)$ $\mathcal{L}(p_\phi(\tilde{s}_i), y_i)$	✗	✗	✓	✗	✓
DAGPromt (WWW 2025 [4])	✗	✓	✗	GLoRA: Θ_{glora} class prompts: $P^{(l)}$ hop coeff.: $\gamma^{(l)}$	$H^{(l)} \leftarrow (A + P_A Q_A^T) H^{(l-1)} (W_0 + P Q^T)$ $S^{(l)} \leftarrow \text{Sim}(H^{(l)}, P^{(l)})$	$\tilde{S} \leftarrow \sum_{l=0}^L \gamma^{(l)} S^{(l)}$ Similarity Loss	✓	✗	✓	✓	✗
ULTRA-DP (arXiv [3])	✗	✓	✗	prompt token: $p_i = p_i^{\text{task}} + w_i^{\text{pos}} p_i^{\text{pos}}$, p_i^{pos} denotes v_i 's positional embedding	create a virtual node v_i^p for target node v_i , $G' \leftarrow (V \cup \{v_i^p\},$ $\mathcal{E} \cup \{(v_i^p, v_i)\}, X \cup \{p_i\})$	Multi-task-Learning	✓	✗	✗	✗	✓
IGAP (WWW 2024 [138])	✗	✓	✓	signal prompts: $\mathcal{P}_s = [p_{s,1}, \dots, p_{s,L}]$ spectral aligner label prompt: $P_l = [p_{l,1}, \dots, p_{l,d}]$	$\tilde{x}_i \leftarrow x_i + \sum_s \alpha_{is} p_s$ $U_K^{pt} \leftarrow P U_K^{ft}$	$\min_{P_s, P_l, \phi}$ $\mathcal{L}_{\text{InfoNCE}}^{cls}$	✓	✗	✓	✓	✗
Self-Pro (ECML PKDD 2024 [22])	✓	✗	✗	self adapter: g_ϕ semantic prompt: s_w structural prompt: ΔA	$t_v \leftarrow \mu s_w + (1 - \mu) h_v$ $t_v^* \leftarrow g_\phi(t_v)$	$L_{NC} = - \sum_{v_i \in V_L} \log$ $\frac{\exp(\text{sim}(t_v^*, t_{v_i}^*) / \tau)}{\sum_{c \in C} \exp(\text{sim}(t_v^*, t_{c_i}^*) / \tau)}$ $L_{LP} = - \sum_{(v_i, a, b) \in \mathcal{T}} \log$ $\frac{\exp(\text{sim}(t_{v_i}^*, t_a^*) / \tau)}{\exp(\text{sim}(t_{v_i}^*, t_a^*) / \tau) + \exp(\text{sim}(t_{v_i}^*, t_b^*) / \tau)}$	✓	✓	✗	✓	✗
PSP (ECML PKDD 2024 [21])	✓	✗	✗	prototype nodes: $\mathcal{P} = \{p_c\}_{c=1}^C$ weighted edges: $W \in \mathbb{R}^{N \times C}$	$G' \leftarrow (V \cup \mathcal{P}, E \cup W)$ $P \leftarrow \text{GNN}([X, X_P], [A, W])$	$L_{pro} = - \frac{1}{ \mathcal{D} } \sum_{(v_i, c_i) \in \mathcal{D}} \log$ $\frac{\exp(\text{sim}(z_i^{(1)}, p_{c_i}) / \tau)}{\sum_{c \in C} \exp(\text{sim}(z_i^{(1)}, p_c) / \tau)}$	✓	✗	✓	✓	✗
SUPT (arXiv [55])	✓	✓	✓	basis prompts: $\{b_j\}_{j=1}^k$ $\alpha^{(m)} = (\bar{D}^{-\frac{1}{2}} \bar{A} \bar{D}^{-\frac{1}{2}})^m (X \oplus \sum_j b_j) W$	$p_i \leftarrow \sum_j \sigma(\alpha)_{ij} b_j$ $X^* \leftarrow X + p$	$\max_{\psi, \phi} P_{f_{\psi, h_\phi}}(y g_\phi(G))$ (+ L_{LP} for SUPTsoft)	✓	✗	✓	✗	✓
EdgePrompt (ICLR 2025 [17])	✓	✓	✓	edge prompts: $\{e_{ij}^{(l)}\}$ anchor prompts: $P^{(l)}$	$h_i^{(l)} \leftarrow \text{COMB}^{(l)}(h_i^{(l-1)},$ $\text{AGG}^{(l)}(\{h_j^{(l-1)}, e_{ij}^{(l)}\}))$	$\min_{g, (P^{(l)}, W^{(l)})} \frac{1}{ V_L } \sum_{v_i \in V_L}$ $\ell_D(g(f(G')_i), y_i)$	✓	✗	✓	✗	✓
UniPromt (NeurIPS 2025 [42])	✓	✓	✓	prompt graph: $\tilde{A}^{init}$ edge weights: $\Psi = \{w_{ij}\}$	$(\tilde{A}^{init})_{ij} \leftarrow S_{ij}, S_{ij} \in \text{top-}k\{S_i\}$ $\tilde{A}^{(l)} \leftarrow \tau \tilde{A}^{(l-1)} + (1 - \tau) \tilde{A}$	$\min_{\phi, \Psi} \frac{1}{ V_L } \sum_{v_i \in V_L}$ $\ell_D(g_\phi(f_{\Psi}(A, X))_i, y_i)$	✓	✗	✗	✗	✓
ProNoG (KDD 2025 [148])	✓	✗	✗	condition-net: CondNet($\cdot; \phi_h$) node prompt: $p_{t,v}$	$s_u \leftarrow \text{Avg}(\{h_u \cdot \text{sim}(h_u, h_v) : u \in V(S_v)\})$ $\tilde{h}_{t,v} \leftarrow p_{t,v} \odot h_v$	$L_{\text{down}} = - \sum_{(x_i, y_i) \in D_t} \log$ $\frac{\exp(\text{sim}(\tilde{h}_{t,x_i}, \tilde{h}_{t,y_i}) / \tau)}{\sum_{c \in \mathcal{Y}} \exp(\text{sim}(\tilde{h}_{t,x_i}, \tilde{h}_{t,c}) / \tau)}$	✓	✗	✓	✓	✗
VNT (KDD 2023 [106])	✓	✓	✗	$P = [p_1; \dots; p_P],$ $p_p \in \mathbb{R}^F$	$[E^1 \ Z^1] = L^1([E^0 \ P])$ $\in \mathbb{R}^{(V+P) \times F}$	Cross Entropy	✓	✗	✗	✗	✓

prompts during message passing. ProNoG [148] targets non-homophilic graphs by generating node-specific prompts from similar neighbors in the local subgraph.

B. Prompt as Graphs. Some methods represent the prompt as an explicit graph rather than as isolated feature vectors. In All in One [103], the prompt is a learnable subgraph whose nodes have the same representation dimension as the original graph nodes. Its token structure consists of links among prompt nodes and cross-links between prompt nodes and original nodes, with link weights computed from feature similarity. The prompted graph is formed by attaching this prompt subgraph to the input graph and then passing the combined graph to the pre-trained encoder.

PRODIGY [41] uses a different graph-form prompt to support graph in-context learning. Its prompt construction contains data graphs and a task graph. Data graphs correspond to task instances, such as subgraphs around target nodes, node pairs, or graph classification examples. The task graph connects data tokens and label tokens, and pre-training is formulated through objectives such as neighboring matching and label matching. This design uses the prompt graph to provide a unified interface for upstream and downstream tasks. PSP [21] also attaches class prototype nodes to the original graph through learnable weighted links, and aligns the prompted graph with a node-level contrastive objective by comparing MLP-encoded node features with GNN-encoded

prompted representations.

Later graph-form prompts focus more on selecting, aligning, or transforming the prompt structure itself. Graph-Prompter [73] maintains a cache of candidate prompt sub-graphs and selects relevant ones through a learned task-graph selector. IGAP [138] instead works in the spectral domain, using signal prompts and a spectral aligner to bridge the pre-training and fine-tuning graph subspaces.

5.2 Aligning Tasks by Answering Function

How do they reformulate downstream tasks to the pretext?

Once the prompt has been inserted, the next issue is how the prompted graph produces an answer. Most methods either reshape downstream instances to match a pre-training objective or add an answering function on top of the prompted representation.

A. Handling Different Level Tasks. All in One [103] pre-trains the graph model with graph-level contrastive learning, where the encoder learns from different views generated by perturbing the original graph. To connect node-, edge-, and graph-level downstream tasks with this graph-level pretext, it converts node and edge instances into induced subgraphs, a strategy also used in related task-reformulation methods [41, 68]. From this view, a graph prompt can be interpreted as a learnable graph modification that approximates operations such as feature masking, edge perturbation, or subgraph transformation. The practical goal is not to claim that a prompt can simulate every graph operation, but to make the downstream instance closer to the representation interface learned during pre-training.

B. Learnable Answering Function. To produce downstream predictions, Sun et al. [103] consider two types of answering functions. The first type is a learnable task head, such as an MLP, that maps the prompted graph representation to the downstream label space. For example, in a three-class node classification task, the target node can first be converted into an induced graph, combined with a graph prompt, encoded by the pre-trained model, and then classified by a three-output prediction head. In this setting, both the prompt and the task head may be tuned. Similar learnable answering functions are also adopted in [12, 13, 106]. These task heads make prediction flexible, but they also introduce an additional tuned component beyond the prompt itself.

C. Hand-crafted Answering Function. The second type uses a preset answering rule without adding a trainable task head. In All in One [103], a node classification task can be handled by constructing K label-specific sub-prompts, where K is the number of classes, and assigning the target node to the class whose prompted graph is most similar to the target instance under a graph-level contrastive objective such as GraphCL [142]. GPPT [101] and HetGPT [74] use link prediction as the task template: the node label is represented as an additional token, and classification is reformulated as estimating the likelihood of a link between the label token and the target node. PSP [21] compares node representations from two graph views, one produced by node feature encoding and the other by a GNN over the prompted graph. Its class prompt tokens are used as reference representations, and prediction is based on similarity to these class tokens. PRODIGY [41] uses a task graph to describe node-, edge-, and graph-level classification through links between data

tokens and label tokens. GraphPrompt [68] extends link prediction to graph-pair similarity and aligns node and graph classification with this pre-training interface. Given induced graphs (g_1, g_2, g_3) where g_1 and g_2 share the same label and g_1 and g_3 have different labels, the answering template encourages $\text{sim}(g_1, g_2) > \text{sim}(g_1, g_3)$. For node classification, the induced graph is the contextual subgraph around the target node.

5.3 Prompt Tuning

How do they learn a graph prompt?

The answering function determines how predictions are read out, but the prompt itself still has to be learned. Existing methods differ mainly in whether they learn prompts across tasks, for a specific downstream task, or under an objective aligned with the pretext task.

A. Meta-Learning Technique. To learn appropriate prompts, Sun et al. [103] leverage meta-learning techniques (such as MAML[16] model) to obtain a robust starting point for the prompt parameters. Since the support set and query set include various graph tasks (such as node classification, link prediction, graph classification, etc), the learned graph prompt is expected to be more general on various downstream tasks.

B. Task-specific Tuning. Besides All in One [103], which aims to learn a general prompt on various downstream tasks, there are also some works that target specific downstream tasks such as graph classification. In this case, the prompt tuning can be more task-directed. For example, GPF [12] aims at a graph classification task, so it just needs to tune the prompt token p and the task head ϕ by maximizing the likelihood of predicted correct graph labels given the prompted graph representation $\tilde{s}_i$ from the pre-trained graph model π . In this case, the task head tuning and the prompt tuning share the same objectives, which can be formulated by: $\max_{p, \phi} \sum_{(y_i, \tilde{s}_i)} p_{\pi, \phi}(y_i | \tilde{s}_i)$. SUPT [55] extends GPF-Plus to the subgraph level by aggregating information from the m -hop neighborhood when computing prompt assignments, so that prompts can reflect the local graph structure around each node.

C. Tuning in Line with Pretext. Intuitively, prompting aims at reformulating downstream tasks to the pre-training task. Therefore, it would be more natural if the prompt tuning shares the same objective with the pre-training task. As suggested in GraphPrompt [68], the authors use a similar loss function to learn prompts. Similarly, GPPT [101] and VNT [106] adopt the same loss function (Cross-Entropy) as their link prediction and node classification tasks, respectively. HetGPT [74], Self-Pro [22], and PSP [21] use node-level contrastive objectives or closely related self-adaptive losses to learn their prompt tokens because their pre-training tasks are also organized around node-pair or node-label comparison. ULTRA-DP [3] develop two pre-training tasks including edge prediction and neighboring prediction, each of which corresponds to one task embedding. In the pre-training phase, they randomly select a task and then integrate specific task-related embeddings into the prompt tokens. These learnable task embeddings are then trained with the graph model. Newer tuning strategies shift attention from prompt placement to prompt optimization. DAGPrompt [4] combines graph LoRA

with hop-wise similarity aggregation and learns from fused similarity scores across layers. RELIEF [165] treats the entire prompt matrix as a feature prompt and tunes it with hierarchical PPO, using the downstream loss drop as the reward signal.

5.4 Further Discussion

What are their connections, pros and cons?

The preceding subsections describe prompt form, answering function, and tuning strategy separately. In practice, these choices are coupled: a prompt that is easy to insert may have limited expressive power, while a more structured prompt may require stronger assumptions about the downstream graph.

GPF [12] shows the appeal of a minimal prompt: a single learnable vector can be added to node features and reused across different pre-training and downstream tasks. Its simplicity also exposes its main limitation. A prompt shared by all nodes has limited expressiveness and may not generalize well across diverse graph instances. Learning an independent prompt for each node would increase flexibility, but it would also be parameter-inefficient. GPF-Plus [13] addresses this tension by learning K basis vectors and composing node-level prompts from them, which brings it closer in spirit to the prompt-token set used in All in One [103].

HetGPT [74] extends prompt tokens to heterogeneous graphs by adding type-specific feature tokens, but its task formulation mainly targets node classification. GraphPrompt [68] takes a different route by using prompt tokens to steer the readout over induced subgraphs and by reformulating node and graph classification through graph-pair similarity. This prompt-guided readout is related in spirit to projection or attention-based graph pooling, but its purpose is different: the prompt is introduced to align downstream prediction with the pre-training interface rather than only to summarize node embeddings.

GPPT [101] can be viewed as a specific instance of the broader All in One framework [103]. If the prompt graph in All in One is reduced to isolated tokens associated with node classes and the resulting graphs are treated as complete graphs, its structure becomes close to the GPPT format. This makes it possible to reuse edge-level pretexts for node classification. The limitation is scope: GPPT is tied to binary edge prediction pretexts and mainly serves node classification. GraphPrompt and All in One cover a wider range of graph tasks, but GraphPrompt, GPF, and GPF-Plus often require an additional task-specific module during adaptation. All in One and GPPT instead rely more on task templates and input manipulation, which reduces their dependence on the form of the downstream classifier.

PRODIGY [41] connects these task-template methods to graph in-context learning. Its data graphs are close to the induced graphs used in All in One and GraphPrompt, while its pre-training task, predicting the link between data tokens and label tokens, follows the same general intuition as GPPT. The main advantage is efficiency: the prompt is fixed and does not require downstream tuning, which makes it suitable for transferring knowledge across datasets. The limitation is flexibility. When the downstream domain differs substantially from the pre-training domain, a non-tunable prompt may not provide enough adaptation capacity. ULTRA-DP [3] takes

almost the opposite route by tuning prompts during both pre-training and downstream adaptation. Its task embeddings are first learned in pre-training and then used to initialize downstream prompts. In this case, the prompt does not simply reformulate the downstream task as a pretext task; it also helps select suitable pre-training tasks from a task pool. This contrast leaves a central trade-off: prompt methods need enough flexibility to adapt, but too much tuning can weaken their efficiency and generality.

VNT [106] sits between explicit prompt insertion and model-internal prompt interaction. Instead of defining prompt-to-node links by hand, it concatenates prompt tokens with the original node set and lets the graph transformer compute their correlations through self-attention. This avoids a manually chosen connection threshold, but it also ties the method to transformer backbones. Since the prompt tokens have no explicit links to the original graph, it can also be harder to combine VNT with graph transformer variants that rely on positional or structural encodings.

This shift explains why recent methods no longer fit neatly into the original token/graph dichotomy. GPF and GPF-Plus are easy to apply because they only modify node features, but they treat graph structure mostly as something handled by the frozen encoder. Self-Pro [22], PSP [21], SUPT [55], and EdgePrompt [17] move further toward structure-aware prompting: they make prompts depend on graph semantics, class prototypes, local neighborhoods, or edges. This gives the prompt more access to graph structure, but it also makes the design less plug-and-play. The method may now depend on neighborhood construction, prototype quality, message-passing layers, or reliable structural signals.

A similar transition appears in prompt optimization. Earlier methods mainly ask whether downstream tasks can be reformulated to match a pretext task. DAGPrompt [4], IGAP [138], RELIEF [165], GraphPrompter [73], and UniPrompt [42] instead focus on what happens when the downstream graph distribution differs from pre-training. DAGPrompt adapts prompts across hops and distributions, IGAP uses spectral alignment to reduce feature and structure gaps, RELIEF learns where and how strongly to prompt through reinforcement learning, GraphPrompter optimizes prompt selection for graph in-context learning, and UniPrompt studies universal adaptation across pre-trained models and domains. These methods are more flexible than early task-template methods, but their gains come with extra assumptions or cost: spectral decompositions may not scale easily, RL-based prompt search adds training complexity, graph in-context methods rely on prompt retrieval quality, and universal adaptation still depends on how broad the pre-training distribution is. These trade-offs motivate the specialized settings discussed next, where graph type, temporal dynamics, modality, and domain shift all affect what a useful prompt should look like.

5.5 Extensions and Specialized Settings

The methods above mainly target general graph prompting. Recent work also studies settings where the graph itself has additional constraints, such as heterogeneous node types, signed edges, or temporal dynamics.

Heterogeneous Graph Prompting. Heterogeneous graphs require prompts to account for node types, edge types, and

relation-specific semantics. HetGPT [74] adds type-specific feature tokens to prompt tokens. HGPROPT [145] bridges homogeneous and heterogeneous graph prompting by aligning structural patterns across relation types. HePa [43] handles node-, edge-, and graph-level classification on heterogeneous graphs. HGMP [47] treats heterogeneous graph prompting as a multi-task problem with contrastive pre-training. MultiHGPT [122] uses task-specific prompts for multi-task heterogeneous graph learning. CLEAR [113] introduces cluster prompts and injects them into meta-paths, which brings high-order heterogeneous semantics into the prompt design.

Signed and Temporal Graphs. Signed graphs carry both positive and negative edges, which requires changing the prompting objective. SGPT [149] adapts prompt tuning to signed graphs under few-shot settings. STGP [38] uses prompts for spatio-temporal graph transfer, and Event-Aware Prompt Learning [144] applies prompts to event-driven dynamic graphs.

6 Graph Prompting in Multi-Modal and Multi-Domain Areas

6.1 Multi-Modal Prompting with Graphs

This section turns to settings in which graph prompting must interact with other modalities or transfer across domains. We discuss two adjacent directions: integrating graphs with modalities such as text, and adapting prompted graph models under domain shift. Current evidence is strongest in text-attributed graphs, textual graph understanding, and molecular graph-language modeling; more general multimodal graph prompting is still less settled. Recent surveys review graph-LLM systems and graph learning under distribution shifts [87, 130, 93]. Here we focus on representative methods in which prompts or prompt-like adaptation interfaces play a central role.

A. Prompting in Text-Attributed Graphs. Early work in text-attributed graphs (TAGs) mainly used prompts to expose graph context to language models or to align text and topology under limited supervision. G2P2 [125] combines graph-grounded pre-training with prompting for low-resource text classification. GIMLET [161] couples graph and text encoders for prompt-based molecular tasks, while Li and Hooi [60] use hand-crafted language templates to inject both raw text and graph structure into node classification. P2TAG [162] makes this coupling more explicit by using mixed prompts that combine text prompts with a prompt graph initialized from label text embeddings. STAGE [172] uses pre-trained LLM embeddings as a lightweight interface for TAGs. GraphiT [51] further studies prompt-optimized LLMs for efficient node classification, and Zhu et al. [166] treat the LLM itself as a TAG foundation model with a learned graph vocabulary. The trend is no longer just to write better textual templates; newer methods try to make the graph structure and text prompt share the same adaptation interface.

B. Large Language Models in Graph Data Processing. Chen et al. [6] study two early pipelines for graph-enhanced node classification: using LLMs to enrich text attributes before GNN prediction, and using LLMs directly as predictors. Fatemi et al. [15] then show that graph serialization and prompt design substantially affect how well

LLMs reason over graph inputs. PATTON [48] pushes this direction toward pre-training on text-rich networks, while KGP [124] uses a knowledge-graph-guided prompting pipeline for multi-document question answering. In the same vein, G-Retriever [33] treats textual graph question answering as a retrieval-augmented generation problem and uses graph-aware retrieval to expose the most relevant substructures to the language model. Relative to earlier graph-serialization studies, this line has shifted from asking whether LLMs can process graph information at all to designing better retrieval, serialization, and prompting interfaces for graph-conditioned reasoning.

C. Multi-modal Fusion with Graph and Prompting. Beyond text, prompt-based multi-modal fusion with graphs is most visible in vision-language and molecular applications. SynerGPT [11] applies in-context prompting to drug synergy prediction. GraphAdapter [59] introduces a graph-aware adapter mechanism for vision-language models, and GIT-Mol [66] jointly models molecular graphs, images, and text with a prompt-driven LLM backbone. HeGraphAdapter [164] studies graph-conditioned adaptation for multi-modal vision-language models with heterogeneous graph structure, and GraphT5 [53] proposes a graph-language model for molecular data through cross-token attention. Current evidence is still concentrated in a few vertical domains, so a reusable recipe for graph-centric multimodal prompting has not yet stabilized.

6.2 Graph Domain Adaptation with Prompting

Graph domain adaptation remains difficult for two reasons: models must align feature semantics across domains, and they must also handle structural shift.

A. Semantic Alignment. Prompt-based graph frameworks provide one route to semantic alignment. All in One [103] provides a prompt-based interface that can be reused across different graph tasks and datasets. GraphControl [170] conditions a pre-trained graph model on downstream-specific information, and Zhang et al. [156] use triple-style prompts for knowledge graph transfer. Yi et al. [141] combine personalized prompts with contrastive learning for cross-domain recommendation, while Liu et al. [62] map graph features into language space and then rely on an LLM encoder to reduce semantic mismatch across domains. More recent cross-domain prompting methods make the adaptation mechanism explicit. UniPrompt [42] maintains learnable edge weights over a prompt graph to adapt pre-trained GNNs to new domains without task-specific tuning. GP2F [31] fuses multiple pre-trained GNNs with adaptive prompts for cross-domain transfer. Collaborate to Adapt [158] addresses source-free graph domain adaptation, and multi-domain graph foundation models suggest that transfer improves when pre-training already exposes the model to semantically diverse graph collections [119, 146]. A practical limitation remains: language-based alignment works best when graph features have interpretable semantics, which is often not the case for dense latent features.

B. Structural Alignment. Cao et al. [2] show that structural gap can place an upper bound on graph transferability, which explains why semantic alignment alone is often insufficient. GraphGLOW [163] tackles this problem by learning graph-shared structural patterns that can be reused across

datasets. DeepGPT [96] adapts graph transformers with prompt tokens at the input and layer levels, which provides a parameter-efficient way to absorb structural differences across datasets. AAGOD [24] is closer to distribution-shift detection than classical domain adaptation, but it is still relevant because it treats a learnable perturbation on the adjacency matrix as a prompt-like mechanism. The same concern has also pushed the field toward multi-graph pre-training and stricter evaluation. Zhao et al. [160] and Wang et al. [119] both study transfer across multiple graph datasets, while Liu et al. [64] show that many reported gains in unsupervised graph domain adaptation remain sensitive to evaluation protocol and benchmark choice. Prompt-based structural adaptation is useful, but its benefit still depends on the diversity of structures seen during pre-training and on how domain shift is measured.

7 Potential Applications

With the widespread utilization of networks as a data modeling structure for representing diverse relational information across social, natural, and academic domains, graph prompting has begun to move from general-purpose adaptation techniques toward concrete application scenarios. In this section, we primarily focus on applications where prompts are instantiated over graph-structured data or graph representations, while also noting related settings where graph knowledge supports prompt-based reasoning. We discuss its emerging applications in online social networks, recommender systems, knowledge management, and biology, with an emphasis on dynamic interaction graphs, user-item graphs, knowledge graphs, and biomedical graphs.

Online Social Networks. Online social platforms consist of users who can be represented as nodes, and their social connections, reposting traces, or interaction events form online social networks (OSNs). Previous research has investigated the potential of prompting mechanisms in identifying fake news within OSNs to prevent malicious attacks. For example, Wu et al. [128] apply textual prompts to pre-trained language models (PLMs) to distill general semantic information, which is then combined with propagation dynamics in social networks for improved classification. Beyond text-oriented prompting, OSNs can also be viewed as dynamic interaction graphs whose node states and edges evolve over time. Chen et al. [5] show that prompt learning can be used to adapt temporal interaction representations to downstream tasks, and DyGPrompt [147] further designs node-time conditional prompts to bridge both task-objective gaps and temporal-distribution gaps between dynamic graph pre-training and downstream inference. These advances make OSN applications less speculative: tasks such as few-shot fake news detection and adaptation over evolving interaction graphs can benefit from prompts that encode temporal and structural context when labeled data are scarce and expensive to obtain.

Recommender Systems. E-commerce platforms provide a valuable opportunity to leverage recommender systems for enhancing online services, because user-item interactions naturally form bipartite or heterogeneous graphs. Early graph prompt tuning studies have explored cross-domain recommendation and contrastive pre-training adaptation [132, 27]. For cross-domain recommendation, Yi et al. [141] introduce extra

prompt nodes when transferring a pre-trained recommendation model to a target domain, thereby improving domain adaptation with limited target supervision. Meanwhile, Yang et al. [139] propose personalized user prompts to bridge the gap between contrastive pretext tasks [127, 143] and downstream recommendation objectives. Recent work further extends this line from static adaptation to dynamic and streaming user-item graphs. GraphPro [140] combines graph pre-training with temporal and structural prompts to improve dynamic recommendation, while GPT4Rec [154] introduces node-level, structure-level, and view-level prompts to guide streaming recommendation over evolving interaction graphs. These studies suggest that graph prompting is particularly suitable for recommender systems where user interests, item popularity, and interaction patterns change continuously, and where efficient adaptation is required for cold-start, cross-domain, and continual recommendation settings.

Knowledge Management. Knowledge graphs (KGs) provide another application scenario in which prompt tuning can be applied directly to graph-structured knowledge. A central line of work performs prompting on KGs using pre-trained KG models to facilitate knowledge transfer across different KG data and tasks. In this line, Zhang et al. [156] design structure-aware pre-training objectives and then employ prompt tuning to transfer task-relevant knowledge to downstream KG tasks. Recent KG completion studies further suggest that adaptive or dual prompting can help align structured relational patterns with task-specific prediction objectives: APKGC [75] uses adaptive prompting for KG completion, and FFCL-KGC [168] incorporates dual prompting into feature-fusion contrastive learning. In parallel, a related but broader direction combines grounded knowledge from KGs with the reasoning and language-generation abilities of LLMs [124, 88, 80, 155, 79]. For example, Tian et al. [108] introduce graph neural prompting to adapt KG knowledge to LLMs in a time- and parameter-efficient manner. These studies indicate that graph prompting can support structured knowledge transfer within KGs, while KG-grounded prompting for LLMs provides a neighboring direction for graph-enhanced knowledge reasoning.

Biology. Molecules can be represented naturally as graphs, where atoms serve as nodes and chemical bonds act as edges [89]. Such graph modeling provides a basis for applying graph representation learning methods to tasks such as molecular property prediction, drug interaction analysis, and protein modeling [89, 69, 161, 11]. Earlier molecular graph learning methods often trained task-specific models on individual datasets [89, 99], which limited cross-task generalization. Although LLMs and language models have been explored as general-purpose tools for understanding molecules [84, 62], text-only descriptions may overlook the internal graph structures of molecules [159]. Recent graph-language models attempt to preserve both structural dependencies and semantic knowledge under few-shot or zero-shot settings [161, 11]. For instance, GIT-Mol [66] uses LoRA [37] for efficient downstream adaptation in molecular LLM systems. Graph prompting now provides more direct biomedical evidence. MOAT [70] designs atom-, geometry-, and task-level prompts for 3D molecular graphs; DDIPrompt [123] applies graph prompt learning to rare and imbalanced drug-drug interaction event prediction; and Prompt-DDG [129] uses microenvironment-

aware hierarchical prompts to predict mutation effects in protein-protein interactions. Together with protein multimer prompt learning [20], these studies show that graph prompting can support efficient task adaptation across molecular, drug-safety, and protein-interaction applications.

8 ProG: Benchmark and Toolkit Support for Graph Prompting

As graph prompting methods diversify, reproducible comparison becomes as important as the design of individual prompts. A method may differ from another one in its pre-training objective, prompt form, downstream task, backbone, shot setting, or evaluation metric, and these choices are often entangled in reported results. ProG was introduced as a graph prompt learning benchmark to reduce this friction by organizing graph prompting experiments around a common pre-train, prompt-tune, and evaluate workflow [171]. In this survey, we treat ProG as part of the supporting infrastructure for the field rather than as another prompt method.

ProG-V2³ extends this benchmark into a more modular and reproducible toolkit. Its architecture, shown in Figure 8, separates graph datasets, pre-training methods, prompt strategies, GNN backbones, downstream tasks, and evaluation reports into explicit components. The current implementation registers prompt strategies through a unified strategy interface and supports representative prompt families such as GPF, GPF-Plus, GraphPrompt, All in One, GPPT, PRODIGY, GraphPrompter, EdgePrompt, RELIEF, UniPrompt, Self-Pro, ProNoG, PSP, and DAGPromptT. It also provides several GNN backbones, including GCN, GAT, GIN, GraphSAGE, GCov, and GraphTransformer, so that prompt behavior can be studied without binding the toolkit to a single encoder.

The toolkit also broadens the evaluation scope beyond a single classification setting. It supports node classification, graph classification, and link prediction through separate task interfaces, and it includes pre-training options such as DGI, GraphMAE, GraphCL, SimGRACE, Edgepred-GPPT, Edgepred-GraphPrompt, and MultiGprompt. The public reports cover a broad set of method combinations, with node- and graph-classification results under few-shot settings and link-prediction results under edge-task metrics such as AUROC and AUPRC. These results are released in a consistent matrix format, making it easier to inspect how different pre-training and prompting choices behave under shared settings.

From an engineering perspective, the V2 version keeps the implementation organized around reusable extension points. Prompt strategies, GNN backbones, task interfaces, configuration, and result export are separated into modules, so new methods can be added without rewriting the full pipeline. This organization supports reproducible comparison while keeping the toolkit practical for researchers who want to test new prompt designs.

More broadly, ProG and curated resources such as Awesome Graph Prompt⁴ complement the taxonomy developed in this survey. The taxonomy helps readers understand how prompt tokens, structures, insertion patterns, answering

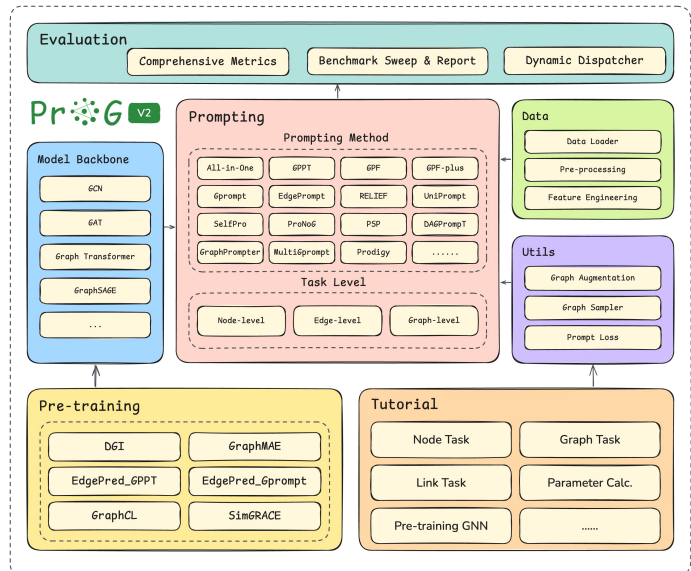


Fig. 8: The architecture of ProG-V2.

functions, and tuning objectives relate to one another. The toolkit provides a practical route for implementing and comparing these choices under shared experimental conventions. As graph prompting moves toward heterogeneous graphs, dynamic graphs, link prediction, and cross-domain transfer, this kind of benchmark infrastructure will be important for separating genuine progress from differences caused by incompatible experimental setups.

9 Challenges and Future Directions

9.1 Current Challenges

Graph prompt learning has made significant research progress, moving from general benchmark tasks to more specialized settings and application scenarios. Nevertheless, the field still faces several challenges in model foundations, evaluation, applications, and transferability. In this subsection, we discuss these challenges in detail.

Inherent Limitation within Graph Models Prompts originate from the NLP field [1, 56, 65], where pre-trained language models provide a relatively unified backbone and textual interface for adapting to downstream tasks. Graph prompt learning faces a different condition. Many graph prompts are not in-context prompts in the same sense as LLM prompts; they are often prompt tokens, prompt graphs, task templates, or structural adapters inserted into graph models. The main limitation is therefore not only the smaller scale of graph models, but also the lack of broadly reusable graph foundation models, unified graph-task interfaces, and sufficiently diverse pre-training corpora across domains. This makes it difficult to design prompts that can consistently activate transferable structural knowledge in the same way that textual prompts can elicit knowledge from large language models.

Intuitive Evaluation of Graph Prompt Learning In the NLP field, prompts are typically in a discrete textual format [1, 88], allowing for intuitive understanding, comparison, and explanation. Graph prompts, in contrast, are usually latent or model-internal. Early methods use learnable tokens [101, 13, 12, 106, 74] or augmented graphs [103, 41], while

3. <https://github.com/sheldonresearch/ProG>

4. <https://github.com/WxxShirley/Awesome-Graph-Prompt>

newer designs further include semantic prompts, class prototypes, edge prompts, spectral prompts, and prompt selection mechanisms [22, 21, 17, 138, 73]. These diverse forms make graph prompting more expressive, but they also make it harder to inspect what is being encoded by a prompt and how different prompt designs should be compared. As a result, evaluation still relies heavily on downstream accuracy, while intuitive measures of prompt quality, structural relevance, and cross-task robustness remain underdeveloped.

More Downstream Applications Graph prompt learning is still largely evaluated on open-source benchmark tasks such as node and graph classification [26, 136], but its application evidence is no longer limited to these settings. DyGPPrompt [147] extends graph prompting to dynamic interaction graphs, while GraphPro [140] and GPT4Rec [154] apply temporal and multi-level prompts to recommender systems. In knowledge graph completion, APKGC [75] and FFCL-KGC [168] show that adaptive or dual prompts can improve structured relational prediction. Biomedical studies such as MOAT [70], DDIPrompt [123], Prompt-DDG [129], and protein multimer prompting [20] further demonstrate application evidence on molecular, drug-interaction, and protein-interaction graphs. However, this progress is uneven across domains. Recommendation and biomedical applications already provide relatively concrete evidence, whereas social-network security, industrial knowledge management, and other high-stakes settings still lack unified evaluation protocols and deployment-oriented studies. The key challenge is to build domain-specific pre-trained graph models and prompt designs that respect the structural, temporal, and semantic constraints of each application scenario.

Transferable Prompt Designs Transferability has become a central concern in graph prompt learning. Early studies [68, 101, 13, 12, 21] often evaluate pre-training and prompt tuning within the same dataset or closely related tasks. PRODIGY [41] explores transferability within the same domain, and All in One [103] provides empirical results regarding transferability across tasks and domains. More recent work further studies semantic alignment [170], structural alignment [163], self-prompting under heterophily [22], universal prompting [42], cross-model prompt fusion [31], source-free graph domain adaptation [158], and multi-domain graph foundation models [119, 146]. Despite these advances, the field still lacks unified transfer protocols and diagnostic tools for explaining why a prompt succeeds in one graph domain but fails in another. Achieving reliable transferability requires joint consideration of task templates, semantic feature spaces, graph structural patterns, and pre-training data diversity.

9.2 Future Directions

With the above analysis on graph prompt, we summarize future directions as follows:

Learning Knowledge from Graph Foundation Models. Currently, graph models are still often tailored to specific domains or tasks, limiting their generalization abilities across broader domains [63, 170]. A significant step toward graph foundation models has been taken by Liu et al. [62], whose OFA model addresses classification tasks for graphs from various domains by mapping graph features into a language-compatible space. Rather than replacing language-based interfaces, graph

prompting can complement graph foundation models and graph-LLM interfaces by providing structure-aware, task-aware, and parameter-efficient adaptation mechanisms. Just as textual prompts adapt LLMs for diverse applications [88, 124, 1], graph prompts may help graph foundation models expose task-specific knowledge while preserving graph topology and structural semantics. A promising direction is therefore to develop prompt interfaces that can work with both graph-native foundation models and graph-language models.

Transferable Learning. As discussed in Section 9.1, graph prompt learning has begun to address transfer across tasks, domains, and pre-trained models, but a systematic understanding of transferability is still missing. Existing methods reformulate different tasks into unified templates [103, 41], align semantic features [170], or adapt structural patterns [163]. More recent methods further consider universal prompts and cross-model adaptation [42, 31]. Future work should move beyond isolated transfer experiments toward standardized cross-dataset, cross-task, and cross-backbone evaluations. It is also important to characterize when a prompt transfers because it captures task semantics, graph structure, or pre-training distribution bias. Such diagnostics would make transferable graph prompting more reliable for heterogeneous, temporal, multi-domain, and application-specific graphs.

Standardized Evaluation and Real-World Deployment. As noted in Section 9.1, graph prompts are usually latent or model-internal, and their assessment still relies heavily on downstream accuracy on a few benchmark tasks [26, 136]. Two coupled directions deserve more attention. First, the community needs intuitive and standardized evaluation protocols that go beyond accuracy to measure prompt quality, structural relevance, parameter efficiency, and cross-task robustness, so that diverse prompt forms [22, 21, 17, 138] can be compared on a common basis. Second, evidence of real-world deployment is still uneven across domains: recommendation and biomedical settings already provide concrete results [140, 154, 123], whereas high-stakes scenarios such as social-network security and industrial knowledge management lack unified protocols and deployment-oriented studies. Building domain-specific pre-trained graph models, together with evaluation suites that respect the structural, temporal, and semantic constraints of each scenario, is a promising path toward trustworthy and deployable graph prompting.

More Theoretical Foundation. Despite the success of graph prompt learning on various downstream tasks, many methods are still motivated by empirical analogies to prompt tuning in NLP [67, 1, 19, 109, 85, 95]. Recent theoretical efforts have started to explain graph prompting from a graph-specific perspective. Wang et al. [117] analyze graph prompts as data operations that approximate graph transformation operators, and derive error bounds for such operations under graph neural models. This view connects prompt insertion with graph augmentation, task reformulation, and upstream-downstream alignment. Nevertheless, the theoretical foundation remains incomplete. Future analysis should further explain how prompt forms interact with graph isomorphism, homophily and heterophily, structural distribution shifts, prompt capacity, and parameter efficiency. Specialized prompts for temporal, heterogeneous, molecular, or knowledge graphs may also require different assumptions from those used for general

graph classification benchmarks.

More Explainable and Understandable Design. While existing graph prompt learning methods have demonstrated impressive results on various downstream tasks [24, 126, 139], we still lack a clear understanding of what exactly is being learned from the prompts. The black-box learning mode of prompt vectors raises questions about their interpretability and whether we can establish meaningful correspondences between input data and the prompted graph [103]. As prompt designs become more structural, explanations should go beyond visualizing prompt vectors and instead clarify how prompts correspond to nodes, edges, subgraphs, prototypes, temporal patterns, or geometric constraints [22, 21, 17, 138]. These issues are crucial for understanding and interpreting prompts but are still underexplored in graph prompt research. To work towards trustworthy graph prompt learning, it is promising to explore the self-interpretability [8, 157] to enable intuitive explanations of graph prompts. By gaining insights into the learned prompt vectors and structures, we can enhance our understanding of the underlying mechanisms and improve the interpretability of graph prompts. This, in turn, can lead to more effective utilization of prompts for high-stakes applications such as recommendation, biomedical modeling, security, and privacy-sensitive graph learning.

10 Conclusion

This survey reviews graph prompt learning as an emerging adaptation paradigm for more transferable and reusable graph learning systems. We organize the literature around transfer across tasks, domains, and modalities, and relate graph prompting to graph pre-training, graph foundation models, multi-modal graph learning, and practical applications. To support systematic comparison, we summarize graph prompts through a unified framework covering prompt tokens, token structures, and insertion patterns.

Beyond method taxonomy, we review application evidence in online social networks, recommender systems, knowledge management, and biomedical graph learning, and discuss supporting benchmark and toolkit resources such as ProG. The field has made clear progress, but it remains far from mature. Key open problems include reusable graph foundation models, reliable transfer protocols, evaluation beyond downstream accuracy, stronger theory, and more interpretable prompt designs. We hope this survey provides a coherent reference point for future work on graph prompting and general-purpose graph learning.

References

- [1] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal et al., “Language Models Are Few-Shot Learners,” in *NeurIPS*, vol. 33, 2020, pp. 1877–1901.
- [2] Y. Cao, J. Xu, C. Yang, J. Wang, Y. Zhang, C. Wang, L. Chen, and Y. Yang, “When to pre-train graph neural networks? An answer from data generation perspective!” in *KDD*, 2023.
- [3] M. Chen, Z. Liu, C. Liu, J. Li, Q. Mao, and J. Sun, “ULTRA-DP: Unifying Graph Pre-training with Multi-task Graph Dual Prompt,” *arXiv preprint*, 2023.
- [4] Q. Chen, L. Wang, B. Zheng, and G. Song, “DAGPrompt: Pushing the Limits of Graph Prompting with a Distribution-aware Graph Prompt Tuning Approach,” in *Proceedings of the ACM Web Conference 2025*. ACM, 2025, pp. 4346–4358. [Online]. Available: <https://doi.org/10.1145/3696410.3714917>
- [5] X. Chen, S. Zhang, Y. Xiong, X. Wu, J. Zhang, X. Sun, Y. Zhang, F. Zhao, and Y. Kang, “Prompt Learning on Temporal Interaction Graphs,” *arXiv preprint*, 2024. [Online]. Available: <https://arxiv.org/abs/2402.06326>
- [6] Z. Chen, H. Mao, H. Li, W. Jin, H. Wen, X. Wei, S. Wang, D. Yin, W. Fan, H. Liu, and J. Tang, “Exploring the Potential of Large Language Models (LLMs) in Learning on Graphs,” *ACM SIGKDD Explorations Newsletter*, 2024.
- [7] J. Cheng, M. Li, J. Li, and F. Tsung, “Wiener graph deconvolutional network improves graph self-supervised learning,” in *AAAI*, 2023, pp. 7131–7139.
- [8] E. Dai and S. Wang, “Towards Self-Explainable Graph Neural Network,” in *CIKM*, 2021, pp. 302–311.
- [9] M. Defferrard, X. Bresson, and P. Vandergheynst, “Convolutional Neural Networks on Graphs with Fast Localized Spectral Filtering,” in *NeurIPS*, vol. 29, 2016.
- [10] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” *NAACL*, 2019.
- [11] C. Edwards, A. Naik, T. Khot, M. Burke, H. Ji, and T. Hope, “SynerGPT: In-Context Learning for Personalized Drug Synergy Prediction and Drug Design,” *arXiv preprint*, 2023.
- [12] T. Fang, Y. Zhang, Y. Yang, and C. Wang, “Prompt tuning for graph neural networks,” *arXiv preprint*, 2022.
- [13] T. Fang, Y. Zhang, Y. Yang, C. Wang, and L. Chen, “Universal Prompt Tuning for Graph Neural Networks,” in *NeurIPS*, 2023.
- [14] X. Fang, L. Liu, J. Lei, D. He, S. Zhang, J. Zhou, F. Wang, H. Wu, and H. Wang, “Geometry-enhanced molecular representation learning for property prediction,” *Nature Machine Intelligence*, vol. 4, no. 2, pp. 127–134, 2022.
- [15] B. Fatemi, J. Halcrow, and B. Perozzi, “Talk like a graph: Encoding graphs for large language models,” *ICLR*, 2024.
- [16] C. Finn, P. Abbeel, and S. Levine, “Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks,” in *ICML*, 2017.
- [17] X. Fu, Y. He, and J. Li, “Edge Prompt Tuning for Graph Neural Networks,” in *The Thirteenth International Conference on Learning Representations*, 2025. [Online]. Available: <https://dblp.org/rec/conf/iclr/FuHL25>
- [18] X. Fu, Z. Wang, Z. Chen, J. Li, Y. Zhu, Z. Lei, C. Shen, Y. Ye, C. Zhang, and J. Li, “Graph Prompting for Graph Learning Models: Recent Advances and Future Directions,” *KDD*, 2025. [Online]. Available: <https://arxiv.org/abs/2506.08326v1>
- [19] T. Gao, A. Fisch, and D. Chen, “Making Pre-Trained Language Models Better Few-Shot Learners,” in *ACL*, 2021, pp. 3816–3830.
- [20] Z. Gao, X. Sun, Z. Liu, Y. Li, H. Cheng, and J. Li, “Protein Multimer Structure Prediction via Prompt Learning,” in *The Twelfth International Conference*

- on Learning Representations, 2024. [Online]. Available: <https://openreview.net/forum?id=OHpvivXrQr>
- [21] Q. Ge, Z. Zhao, Y. Liu, A. Cheng, X. Li, S. Wang, and D. Yin, “PSP: Pre-Training and Structure Prompt Tuning for Graph Neural Networks,” in *Machine Learning and Knowledge Discovery in Databases. Research Track, ECML PKDD 2024*, 2024, pp. 423–439. [Online]. Available: <https://dblp.org/rec/conf/pkdd/GeZLCLWY24>
 - [22] C. Gong, X. Li, J. Yu, Y. Cheng, J. Tan, and C. Yu, “Self-Pro: A Self-Prompt and Tuning Framework for Graph Neural Networks,” in *Machine Learning and Knowledge Discovery in Databases. Research Track, ECML PKDD 2024*, 2024, pp. 197–215. [Online]. Available: <https://dblp.org/rec/conf/pkdd/GongLYCTY24>
 - [23] A. Grover and J. Leskovec, “node2vec: Scalable feature learning for networks,” in *KDD*, 2016, pp. 855–864.
 - [24] Y. Guo, C. Yang, Y. Chen, J. Liu, C. Shi, and J. Du, “A Data-Centric Framework to Endow Graph Neural Networks with Out-of-Distribution Detection Ability,” in *KDD*, 2023, pp. 638–648.
 - [25] Y. Guo, D. Bo, C. Yang, Z. Lu, Z. Zhang, J. Liu, Y. Peng, and C. Shi, “Data-centric graph learning: A survey,” *IEEE Transactions on Big Data*, vol. 11, no. 1, pp. 1–20, 2025.
 - [26] W. Hamilton, Z. Ying, and J. Leskovec, “Inductive Representation Learning on Large Graphs,” in *NeurIPS*, vol. 30, 2017.
 - [27] B. Hao, C. Yang, L. Guo, J. Yu, and H. Yin, “Motif-Based Prompt Learning for Universal Cross-Domain Recommendation,” in *WSDM*, 2024.
 - [28] A. Hasanzadeh, E. Hajiramezanali, K. Narayanan, N. Duffield, M. Zhou, and X. Qian, “Semi-Implicit Graph Variational Auto-Encoders,” in *NeurIPS*, vol. 32, 2019.
 - [29] K. Hassani and A. H. K. Ahmadi, “Contrastive Multi-View Representation Learning on Graphs,” in *ICML*, vol. 119, 2020, pp. 4116–4126.
 - [30] A. Haviv, J. Berant, and A. Globerson, “BERTese: Learning to Speak to BERT,” in *EACL*, 2021, pp. 3618–3623.
 - [31] D. He, W. Sun, Y. Huang, J. Zhao, and D. Jin, “GP2F: Cross-Domain Graph Prompting with Adaptive Fusion of Pre-trained Graph Neural Networks,” *arXiv preprint*, 2026. [Online]. Available: <https://arxiv.org/abs/2602.11629v1>
 - [32] M. He, Z. Wei, z. Huang, and H. Xu, “BernNet: Learning Arbitrary Graph Spectral Filters via Bernstein Approximation,” in *NeurIPS*, vol. 34, 2021, pp. 14 239–14 251.
 - [33] X. He, Y. Tian, Y. Sun, N. V. Chawla, T. Laurent, Y. LeCun, X. Bresson, and B. Hooi, “G-Retriever: Retrieval-augmented generation for textual graph understanding and question answering,” in *Advances in Neural Information Processing Systems* 37, 2024. [Online]. Available: <https://doi.org/10.52202/079017-4224>
 - [34] Y. He, Y. Sui, X. He, and B. Hooi, “UniGraph: Learning a unified cross-domain foundation model for text-attributed graphs,” in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2025, pp. 448–459. [Online]. Available: <https://doi.org/10.1145/3690624.3709277>
 - [35] Z. Hou, X. Liu, Y. Cen, Y. Dong, H. Yang, C. Wang, and J. Tang, “GraphMAE: Self-Supervised Masked Graph Autoencoders,” in *KDD*, 2022, pp. 594–604.
 - [36] Z. Hou, Y. He, Y. Cen, X. Liu, Y. Dong, E. Kharlamov, and J. Tang, “GraphMAE2: A Decoding-Enhanced Masked Self-Supervised Graph Learner,” in *The Web Conference*, 2023, pp. 737–746.
 - [37] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-Rank Adaptation of Large Language Models,” *ICLR*, 2022.
 - [38] J. Hu, X. Liu, Z. Fan, Y. Yin, S. Xiang, S. Ramasamy, and R. Zimmermann, “Prompt-based spatio-temporal graph transfer learning,” in *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*. ACM, 2024, pp. 890–899. [Online]. Available: <https://doi.org/10.1145/3627673.3679554>
 - [39] W. Hu, B. Liu, J. Gomes, M. Zitnik, P. Liang, V. S. Pande, and J. Leskovec, “Strategies for Pre-training Graph Neural Networks,” in *ICLR*, 2020.
 - [40] Z. Hu, Y. Dong, K. Wang, K.-W. Chang, and Y. Sun, “GPT-GNN: Generative Pre-Training of Graph Neural Networks,” in *KDD*, 2020, pp. 1857–1867.
 - [41] Q. Huang, H. Ren, P. Chen, G. Kržmanc, D. Zeng, P. Liang, and J. Leskovec, “PRODIGY: Enabling In-context Learning Over Graphs,” in *NeurIPS*, 2023.
 - [42] Y. Huang, J. Zhao, D. He, X. Wang, Y. Li, Y. Huang, D. Jin, and Z. Feng, “One Prompt Fits All: Universal Graph Adaptation for Pretrained Models,” in *Advances in Neural Information Processing Systems*, 2025. [Online]. Available: <https://openreview.net/forum?id=xJ2lGfFov7>
 - [43] J. Jia, L. Song, J. Li, and Y. Kong, “HePa: Heterogeneous graph prompting for all-level classification tasks,” in *Proceedings of the AAAI Conference on Artificial Intelligence*. Association for the Advancement of Artificial Intelligence (AAAI), 2025. [Online]. Available: <https://doi.org/10.1609/aaai.v39i11.33297>
 - [44] X. Jiang, T. Jia, Y. Fang, C. Shi, Z. Lin, and H. Wang, “Pre-training on Large-Scale Heterogeneous Graph,” in *KDD*, 2021, pp. 756–766.
 - [45] X. Jiang, Y. Lu, Y. Fang, and C. Shi, “Contrastive Pre-Training of GNNs on Heterogeneous Graphs,” in *CIKM*, 2021, pp. 803–812.
 - [46] Z. Jiang, F. F. Xu, J. Araki, and G. Neubig, “How Can We Know What Language Models Know?” *TACL*, vol. 8, pp. 423–438, 2020.
 - [47] P. Jiao, J. Ni, D. Jin, X. Guo, and H. Liu, “HGMP: Heterogeneous graph multi-task prompt learning,” in *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*. IJCAI, 2025, pp. 2982–2990. [Online]. Available: <https://doi.org/10.24963/ijcai.2025/332>
 - [48] B. Jin, W. Zhang, Y. Zhang, Y. Meng, X. Zhang, Q. Zhu, and J. Han, “Patton: Language Model Pretraining on Text-Rich Networks,” in *ACL*, 2023.
 - [49] M. Jin, Y. Zheng, Y.-F. Li, C. Gong, C. Zhou, and

- S. Pan, “Multi-Scale Contrastive Siamese Networks for Self-Supervised Graph Representation Learning,” in *IJCAI*, 2021, pp. 1477–1483.
- [50] W. Jin, T. Derr, Y. Wang, Y. Ma, Z. Liu, and J. Tang, “Node Similarity Preserving Graph Convolutional Networks,” in *WSDM*, 2021, pp. 148–156.
- [51] S. Khoshraftar, N. Abedini, and A. Hajian, “Graphit: Efficient node classification on text-attributed graphs with prompt optimized llms,” *WWW*, 2025. [Online]. Available: <https://arxiv.org/abs/2502.10522>
- [52] D. Kim and A. Oh, “How to Find Your Friendly Neighborhood: Graph Attention Design with Self-Supervision,” in *ICLR*, 2021.
- [53] S. Kim, N. Kim, Y. Piao, and S. Kim, “Graph5: Unified molecular graph-language modeling via multi-modal cross-token attention,” *arXiv preprint*, 2025. [Online]. Available: <https://arxiv.org/abs/2503.07655>
- [54] D. Lachi, M. Azabou, V. Arora, and E. Dyer, “GraphFM: A generalist graph transformer that learns transferable representations across diverse domains,” *Transactions on Machine Learning Research*, 2025. [Online]. Available: <https://openreview.net/forum?id=sZTpRfRUtR>
- [55] J. Lee, W. Yang, and J. Kang, “Subgraph-level Universal Prompt Tuning,” *CoRR*, vol. abs/2402.10380, 2024. [Online]. Available: <https://arxiv.org/abs/2402.10380>
- [56] B. Lester, R. Al-Rfou, and N. Constant, “The Power of Scale for Parameter-Efficient Prompt Tuning,” in *EMNLP*, 2021, pp. 3045–3059.
- [57] J. Li, R. Wu, W. Sun, L. Chen, S. Tian, L. Zhu, C. Meng, Z. Zheng, and W. Wang, “What’s Behind the Mask: Understanding Masked Graph Modeling for Graph Autoencoders,” in *KDD*, 2023, pp. 1268–1279.
- [58] X. L. Li and P. Liang, “Prefix-Tuning: Optimizing Continuous Prompts for Generation,” in *ACL-IJCNLP*, 2021, pp. 4582–4597.
- [59] X. Li, D. Lian, Z. Lu, J. Bai, Z. Chen, and X. Wang, “GraphAdapter: Tuning Vision-Language Models With Dual Knowledge Graph,” in *NeurIPS*, 2023.
- [60] Y. Li and B. Hooi, “Prompt-Based Zero-and Few-Shot Node Classification: A Multimodal Approach,” *arXiv preprint*, 2023.
- [61] Y. Li, Z. Li, P. Wang, J. Li, X. Sun, H. Cheng, and J. X. Yu, “A survey of graph meets large language model: Progress and future directions,” *IJCAI*, 2024.
- [62] H. Liu, J. Feng, L. Kong, N. Liang, D. Tao, Y. Chen, and M. Zhang, “One for All: Towards Training One Graph Model for All Classification Tasks,” *ICLR*, 2024.
- [63] J. Liu, C. Yang, Z. Lu, J. Chen, Y. Li, M. Zhang, T. Bai, Y. Fang, L. Sun, P. S. Yu, and C. Shi, “Towards Graph Foundation Models: A Survey and Beyond,” *arXiv preprint*, 2023.
- [64] M. Liu, Z. Zhang, J. Tang, J. Bu, B. He, and S. Zhou, “Revisiting, benchmarking and understanding unsupervised graph domain adaptation,” in *Advances in Neural Information Processing Systems* 37, 2024. [Online]. Available: <https://doi.org/10.52202/079017-2838>
- [65] P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, and G. Neubig, “Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing,” *ACM Computing Surveys*, vol. 55, no. 9, pp. 195:1–195:35, 2023.
- [66] P. Liu, Y. Ren, J. Tao, and Z. Ren, “GIT-Mol: A Multi-modal Large Language Model for Molecular Science with Graph, Image, and Text,” *Computers in Biology and Medicine*, 2024.
- [67] X. Liu, K. Ji, Y. Fu, W. Tam, Z. Du, Z. Yang, and J. Tang, “P-Tuning: Prompt Tuning Can Be Comparable to Fine-Tuning Across Scales and Tasks,” in *ACL*, vol. 2, 2022, pp. 61–68.
- [68] Z. Liu, X. Yu, Y. Fang, and X. Zhang, “Graph-prompt: Unifying Pre-Training and Downstream Tasks for Graph Neural Networks,” in *The Web Conference*, 2023, pp. 417–428.
- [69] Z. Liu, S. Li, Y. Luo, H. Fei, Y. Cao, K. Kawaguchi, X. Wang, and T.-S. Chua, “MolCA: Molecular Graph-Language Modeling with Cross-Modal Projector and Uni-Modal Adapter,” in *EMNLP*, 2023.
- [70] Q. Long, Y. Yan, W. Cui, W. Ju, Z. Zhu, Y. Zhou, X. Wang, and M. Xiao, “MOAT: Graph prompting for 3d molecular graphs,” in *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management. Association for Computing Machinery*, 2024, pp. 1586–1596. [Online]. Available: <https://doi.org/10.1145/3627673.3679628>
- [71] Q. Long, Y. Yan, P. Zhang, C. Fang, W. Cui, Z. Ning, M. Xiao, N. Cao, X. Luo, L. Xu, S. Jiang, Z. Fang, C. Chen, X.-S. Hua, and Y. Zhou, “Towards Graph Prompt Learning: A Survey and Beyond,” *arXiv preprint*, 2024. [Online]. Available: <https://arxiv.org/abs/2408.14520v3>
- [72] Y. Long, M. Wu, Y. Liu, Y. Fang, C. K. Kwok, J. Chen, J. Luo, and X. Li, “Pre-training graph neural networks for link prediction in biomedical networks,” *Bioinformatics*, vol. 38, no. 8, pp. 2254–2262, 2022.
- [73] R. Lv, Z. Zhang, K. Zhang, Q. Liu, W. Gao, J. Liu, J. Yan, L. Yue, and F. Yao, “Graphprompter: Multi-stage Adaptive Prompt Optimization for Graph In-Context Learning,” in *2025 IEEE 41st International Conference on Data Engineering (ICDE)*. IEEE, 2025, pp. 3917–3930. [Online]. Available: <https://doi.org/10.1109/ICDE65448.2025.00292>
- [74] Y. Ma, N. Yan, J. Li, M. Mortazavi, and N. V. Chawla, “HetGPT: Harnessing the Power of Prompt Tuning in Pre-Trained Heterogeneous Graph Neural Networks,” in *Proceedings of the ACM Web Conference 2024*, 2024.
- [75] A. Maolinyazi, C. Ma, and X.-F. Meng, “APKGC: An adaptive and prompt-tuning framework for knowledge graph completion,” *Knowledge-Based Systems*, vol. 333, p. 114950, 2026. [Online]. Available: <https://doi.org/10.1016/j.knosys.2025.114950>
- [76] M. Niepert, M. Ahmed, and K. Kutzkov, “Learning Convolutional Neural Networks for Graphs,” in *ICML*, 2016, pp. 2014–2023.
- [77] M. Ou, P. Cui, J. Pei, Z. Zhang, and W. Zhu, “Asymmetric Transitivity Preserving Graph Embedding,” in *KDD*, 2016, pp. 1105–1114.
- [78] S. Pan, R. Hu, G. Long, J. Jiang, L. Yao, and C. Zhang, “Adversarially Regularized Graph Autoencoder for Graph Embedding,” in *IJCAI*, 2018, pp. 2609–

- 2615.
- [79] S. Pan, L. Luo, Y. Wang, C. Chen, J. Wang, and X. Wu, “Unifying Large Language Models and Knowledge Graphs: A Roadmap,” *IEEE Transactions on Knowledge and Data Engineering*, 2024.
- [80] J. Park, A. Patel, O. Z. Khan, H. J. Kim, and J.-K. Kim, “Graph-Guided Reasoning for Multi-Hop Question Answering in Large Language Models,” *arXiv preprint*, 2023.
- [81] J. Park, M. Lee, H. J. Chang, K. Lee, and J. Y. Choi, “Symmetric Graph Convolutional Autoencoder for Unsupervised Graph Representation Learning,” in *ICCV*, 2019, pp. 6519–6528.
- [82] Z. Peng, W. Huang, M. Luo, Q. Zheng, Y. Rong, T. Xu, and J. Huang, “Graph Representation Learning via Graphical Mutual Information Maximization,” in *The Web Conference*, 2020, pp. 259–270.
- [83] B. Perozzi, R. Al-Rfou, and S. Skiena, “Deepwalk: Online learning of social representations,” in *KDD*, 2014, pp. 701–710.
- [84] C. Qian, H. Tang, Z. Yang, H. Liang, and Y. Liu, “Can Large Language Models Empower Molecular Property Prediction?” *arXiv preprint*, 2023.
- [85] G. Qin and J. Eisner, “Learning How to Ask: Querying LMs with Mixtures of Soft Prompts,” in *NAACL-HLT*, 2021, pp. 5203–5212.
- [86] J. Qiu, Q. Chen, Y. Dong, J. Zhang, H. Yang, M. Ding, K. Wang, and J. Tang, “GCC: Graph Contrastive Coding for Graph Neural Network Pre-Training,” in *KDD*, 2020, pp. 1150–1160.
- [87] X. Ren, J. Tang, D. Yin, N. Chawla, and C. Huang, “A survey of large language models for graphs,” in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024, pp. 6616–6626. [Online]. Available: <https://doi.org/10.1145/3637528.3671460>
- [88] J. Robinson, C. M. Rytting, and D. Wingate, “Leveraging Large Language Models for Multiple Choice Question Answering,” *ICLR*, 2023.
- [89] Y. Rong, Y. Bian, T. Xu, W. Xie, Y. WEI, W. Huang, and J. Huang, “Self-Supervised Graph Transformer on Large-Scale Molecular Data,” in *NeurIPS*, vol. 33, 2020, pp. 12 559–12 571.
- [90] M. T. Rosenstein, Z. Marx, L. P. Kaelbling, and T. G. Dietterich, “To transfer or not to transfer,” in *NeurIPS*, vol. 898, 2005.
- [91] T. Schick and H. Schütze, “Few-Shot Text Generation with Natural Language Instructions,” in *EMNLP*, 2021, pp. 390–402.
- [92] —, “It’s Not Just Size That Matters: Small Language Models Are Also Few-Shot Learners,” in *NAACL-HLT*, 2021, pp. 2339–2352.
- [93] B. Shi, Y. Wang, F. Guo, B. Xu, H. Shen, and X. Cheng, “Domain adaptation for graph representation learning: Challenges, progress, and prospects,” *Journal of Computer Science and Technology*, vol. 40, no. 2, pp. 283–300, 2025. [Online]. Available: <https://doi.org/10.1007/s11390-024-4465-x>
- [94] Y. Shi, Z. Huang, S. Feng, H. Zhong, W. Wang, and Y. Sun, “Masked Label Prediction: Unified Message Passing Model for Semi-Supervised Classification,” in *IJCAI*, vol. 2, 2021, pp. 1548–1554.
- [95] T. Shin, Y. Razeghi, R. L. L. IV, E. Wallace, and S. Singh, “Autoprompt: Eliciting Knowledge from Language Models with Automatically Generated Prompts,” in *EMNLP*, 2020, pp. 4222–4235.
- [96] R. Shirkavand and H. Huang, “Deep Prompt Tuning for Graph Transformers,” *arXiv preprint*, 2023.
- [97] Y. Song, H. Mao, J. Xiao, J. Liu, Z. Chen, W. Jin, C. Yang, J. Tang, and H. Liu, “A pure transformer pretraining framework on text-attributed graphs,” in *Proceedings of the Third Learning on Graphs Conference*, ser. *Proceedings of Machine Learning Research*, vol. 269. PMLR, 2025, pp. 24:1–24:18. [Online]. Available: <https://proceedings.mlr.press/v269/song25a.html>
- [98] A. Subramonian, “MOTIF-Driven Contrastive Learning of Graph Representations,” in *AAAI*, vol. 35, 2021, pp. 15 980–15 981.
- [99] F.-Y. Sun, J. Hoffmann, V. Verma, and J. Tang, “InfoGraph: Unsupervised and Semi-supervised Graph-Level Representation Learning via Mutual Information Maximization,” in *ICLR*, 2020.
- [100] M. Sun, J. Xing, H. Wang, B. Chen, and J. Zhou, “MoCL: Data-driven Molecular Fingerprint via Knowledge-aware Contrastive Learning from Molecular Graph,” in *KDD*, 2021, pp. 3585–3594.
- [101] M. Sun, K. Zhou, X. He, Y. Wang, and X. Wang, “GPPT: Graph Pre-Training and Prompt Tuning to Generalize Graph Neural Networks,” in *KDD*, 2022, pp. 1717–1727.
- [102] X. Sun, H. Yin, B. Liu, H. Chen, J. Cao, Y. Shao, and N. Q. Viet Hung, “Heterogeneous Hypergraph Embedding for Graph Classification,” in *WSDM*, 2021, pp. 725–733.
- [103] X. Sun, H. Cheng, J. Li, B. Liu, and J. Guan, “All in One: Multi-Task Prompting for Graph Neural Networks,” in *KDD*, 2023, pp. 2120–2131.
- [104] S. Suresh, P. Li, C. Hao, and J. Neville, “Adversarial Graph Augmentation to Improve Graph Contrastive Learning,” in *NeurIPS*, vol. 34, 2021, pp. 15 920–15 933.
- [105] Q. Tan, N. Liu, X. Huang, S.-H. Choi, L. Li, R. Chen, and X. Hu, “S2GAE: Self-Supervised Graph Autoencoders are Generalizable Learners with Graph Masking,” in *Proceedings of the Sixteenth ACM International Conference on Web Search and Data Mining*, ser. *WSDM ’23*, 2023, pp. 787–795. [Online]. Available: <https://doi.org/10.1145/3539597.3570404>
- [106] Z. Tan, R. Guo, K. Ding, and H. Liu, “Virtual Node Tuning for Few-shot Node Classification,” in *KDD*, 2023, pp. 2177–2188.
- [107] S. Thakoor, C. Tallec, M. G. Azar, R. Munos, P. Veličković, and M. Valko, “Bootstrapped representation learning on graphs,” in *ICLR*, 2021.
- [108] Y. Tian, H. Song, Z. Wang, H. Wang, Z. Hu, F. Wang, N. V. Chawla, and P. Xu, “Graph Neural Prompting with Large Language Models,” *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.
- [109] M. Tsimpoukelli, J. L. Menick, S. Cabi, S. M. A. Eslami, O. Vinyals, and F. Hill, “Multimodal Few-Shot Learning with Frozen Language Models,” in *NeurIPS*, vol. 34, 2021, pp. 200–212.

- [110] P. Velickovic, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, “Graph Attention Networks,” in ICLR, 2018.
- [111] P. Velickovic, W. Fedus, W. L. Hamilton, P. Liò, Y. Bengio, and R. D. Hjelm, “Deep Graph Infomax,” in ICLR, 2019.
- [112] C. Wang, S. Pan, G. Long, X. Zhu, and J. Jiang, “MGAE: Marginalized Graph Autoencoder for Graph Clustering,” in CIKM, 2017, pp. 889–898.
- [113] F. Wang, Z. Zhang, J. Ye, L. Sun, and J. Qi, “CLEAR: Cluster-based Prompt Learning on Heterogeneous Graphs,” PAKDD, 2025. [Online]. Available: <https://arxiv.org/abs/2502.08918v1>
- [114] H. Wang, T. Fu, Y. Du, W. Gao, K. Huang, Z. Liu et al., “Scientific discovery in the age of artificial intelligence,” *Nature*, vol. 620, no. 7972, pp. 47–60, 2023.
- [115] J. Wang, D. Chen, C. Luo, X. Dai, L. Yuan, Z. Wu, and Y.-G. Jiang, “ChatVideo: A Tracklet-centric Multimodal and Versatile Video Understanding System,” arXiv preprint, 2023.
- [116] P. Wang, K. Agarwal, C. Ham, S. Choudhury, and C. K. Reddy, “Self-Supervised Learning of Contextual Embeddings for Link Prediction in Heterogeneous Networks,” in The Web Conference, 2021, pp. 2946–2957.
- [117] Q. Wang, X. Sun, and H. Cheng, “Does graph prompt work? a data operation perspective with theoretical analysis,” in Proceedings of the 42nd International Conference on Machine Learning, 2025.
- [118] R. Wang, Y. Li, S. Lin, W. Wu, H. Xie, Y. Xu, and J. C. Lui, “Common neighbors matter: fast random walk sampling with common neighbor awareness,” TKDE, vol. 35, no. 5, pp. 4570–4584, 2022.
- [119] S. Wang, B. Wang, Z. Shen, B. Deng, and Z. Kang, “Multi-domain graph foundation models: Robust knowledge transfer via topology alignment,” ICML, 2025. [Online]. Available: <https://arxiv.org/abs/2502.02017>
- [120] X. Wang, Y. Zhang, and C. Shi, “Hyperbolic heterogeneous information network embedding,” in AAAI, vol. 33, 2019, pp. 5337–5344.
- [121] X. Wang, N. Liu, H. Han, and C. Shi, “Self-supervised Heterogeneous Graph Neural Network with Co-contrastive Learning,” in KDD, 2021, pp. 1726–1736.
- [122] Y. Wang, Y. Cao, Z. Li, B. Dong, G. Ye, and H. Chai, “MultiHGPT: Multi-task heterogeneous graph prompt tuning,” *Information Processing & Management*, vol. 62, no. 6, p. 104236, 2025. [Online]. Available: <https://doi.org/10.1016/j.ipm.2025.104236>
- [123] Y. Wang, Y. Xiong, X. Wu, X. Sun, J. Zhang, and G. Zheng, “DDIPrompt: Drug-drug interaction event prediction based on graph prompt learning,” in Proceedings of the 33rd ACM International Conference on Information and Knowledge Management. Association for Computing Machinery, 2024, pp. 2431–2441. [Online]. Available: <https://doi.org/10.1145/3627673.3679645>
- [124] Y. Wang, N. Lipka, R. A. Rossi, A. Siu, R. Zhang, and T. Derr, “Knowledge Graph Prompting for Multi-Document Question Answering,” Proceedings of the AAAI Conference on Artificial Intelligence, 2024.
- [125] Z. Wen and Y. Fang, “Augmenting Low-Resource Text Classification with Graph-Grounded Pre-Training and Prompting,” in SIGIR, 2023, pp. 506–516.
- [126] Z. Wen, Y. Fang, Y. Liu, Y. Guo, and S. Hao, “Voucher Abuse Detection with Prompt-based Fine-tuning on Graph Neural Networks,” in Proceedings of the 32nd ACM International Conference on Information and Knowledge Management, 2023.
- [127] J. Wu, X. Wang, F. Feng, X. He, L. Chen, J. Lian, and X. Xie, “Self-supervised Graph Learning for Recommendation,” in SIGIR, 2021.
- [128] J. Wu, S. Li, A. Deng, M. Xiong, and H. Bryan, “Prompt-and-Align: Prompt-Based Social Alignment for Few-Shot Fake News Detection,” in CIKM, 2023.
- [129] L. Wu, Y. Tian, H. Lin, Y. Huang, S. Li, N. V. Chawla, and S. Z. Li, “Learning to predict mutational effects of protein-protein interactions by microenvironment-aware hierarchical prompt learning,” in Proceedings of the 41st International Conference on Machine Learning, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 53 847–53 859. [Online]. Available: <https://proceedings.mlr.press/v235/wu24y.html>
- [130] M. Wu, X. Zheng, Q. Zhang, X. Shen, X. Luo, X. Zhu, and S. Pan, “Graph learning under distribution shifts: A comprehensive survey on domain adaptation, out-of-distribution, and continual learning,” arXiv preprint, 2024. [Online]. Available: <https://arxiv.org/abs/2402.16374>
- [131] X. Wu, K. Zhou, M. Sun, X. Wang, and N. Liu, “A Survey of Graph Prompting Methods: Techniques, Applications, and Challenges,” arXiv preprint, 2023.
- [132] Y. Wu, R. Xie, Y. Zhu, F. Zhuang, X. Zhang, L. Lin, and Q. He, “Personalized Prompt for Sequential Recommendation,” *IEEE Transactions on Knowledge and Data Engineering*, 2024.
- [133] Y. Xie, Z. Xu, and S. Ji, “Self-Supervised Representation Learning via Latent Graph Prediction,” in ICML, 2022, pp. 24 460–24 477.
- [134] Y. Xie, Z. Xu, J. Zhang, Z. Wang, and S. Ji, “Self-Supervised Learning of Graph Neural Networks: A Unified Review,” TPAML, vol. 45, no. 2, pp. 2412–2429, 2023.
- [135] D. Xu, W. Cheng, D. Luo, H. Chen, and X. Zhang, “InfoGCL: Information-Aware Graph Contrastive Learning,” in NeurIPS, vol. 34, 2021, pp. 30 414–30 425.
- [136] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, “How Powerful are Graph Neural Networks?” in ICLR, 2019.
- [137] Y. Xu, X. Liu, K. Duan, Y. Fang, Y.-N. Chuang, D. Zha, and Q. Tan, “Graphfm: A comprehensive benchmark for graph foundation model,” arXiv preprint, 2024. [Online]. Available: <https://arxiv.org/abs/2406.08310v2>
- [138] Y. Yan, P. Zhang, Z. Fang, and Q. Long, “Inductive Graph Alignment Prompt: Bridging the Gap between Graph Pre-training and Inductive Fine-tuning From Spectral Perspective,” in Proceedings of the ACM Web Conference 2024. ACM, 2024, pp. 4328–4339. [Online]. Available: <https://doi.org/10.1145/3589334.3645620>
- [139] H. Yang, X. Zhao, Y. Li, H. Chen, and G. Xu, “An Empirical Study Towards Prompt-Tuning for Graph Contrastive Pre-Training in Recommendations,” in NeurIPS, 2023.

- [140] Y. Yang, L. Xia, D. Luo, K. Lin, and C. Huang, “GraphPro: Graph pre-training and prompt learning for recommendation,” in *Proceedings of the ACM Web Conference 2024*. Association for Computing Machinery, 2024, pp. 3690–3699. [Online]. Available: <https://doi.org/10.1145/3589334.3645546>
- [141] Z. Yi, I. Ounis, and C. Macdonald, “Contrastive Graph Prompt-tuning for Cross-domain Recommendation,” *ACM Transactions on Information Systems*, vol. 42, no. 2, pp. 1–28, 2024.
- [142] Y. You, T. Chen, Y. Sui, T. Chen, Z. Wang, and Y. Shen, “Graph contrastive learning with augmentations,” in *NeurIPS*, 2020, pp. 5812–5823.
- [143] J. Yu, H. Yin, X. Xia, T. Chen, L. Cui, and N. Q. V. Hung, “Are Graph Augmentations Necessary? Simple Graph Contrastive Learning for Recommendation,” in *SIGIR*, 2022.
- [144] X. Yu, R. Liang, X. Zhang, and Y. Fang, “Event-aware prompt learning for dynamic graphs,” primary Class: cs.LG. [Online]. Available: <https://arxiv.org/abs/2510.11339>
- [145] X. Yu, Y. Fang, Z. Liu, and X. Zhang, “HGPROMP: Bridging homogeneous and heterogeneous graphs for few-shot prompt learning,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 15. Association for the Advancement of Artificial Intelligence (AAAI), 2024, pp. 16 578–16 586. [Online]. Available: <https://doi.org/10.1609/aaai.v38i15.29596>
- [146] X. Yu, Z. Gong, C. Zhou, Y. Fang, and H. Zhang, “SAMGPT: Text-free graph foundation model for multi-domain pre-training and cross-domain adaptation,” in *Proceedings of the ACM Web Conference 2025*, 2025, pp. 1142–1153. [Online]. Available: <https://doi.org/10.1145/3696410.3714828>
- [147] X. Yu, Z. Liu, X. Zhang, and Y. Fang, “Node-time conditional prompt learning in dynamic graphs,” in *Proceedings of the 13th International Conference on Learning Representations*, 2025. [Online]. Available: <https://openreview.net/forum?id=kVlfYvIqaK>
- [148] X. Yu, J. Zhang, Y. Fang, and R. Jiang, “Non-homophilic Graph Pre-Training and Prompt Learning,” in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2025, pp. 1844–1854. [Online]. Available: <https://dblp.org/rec/conf/kdd/YuZFJ25>
- [149] Z. Zhai, S. Qing, X. Wang, and W. Zhang, “SGPT: Few-Shot Prompt Tuning for Signed Graphs,” *CIKM*, 2025. [Online]. Available: <https://arxiv.org/abs/2412.12155v2>
- [150] H. Zhang, X. Li, and L. Bing, “Video-LLaMA: An Instruction-tuned Audio-Visual Language Model for Video Understanding,” in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 2023.
- [151] J. Zhang, H. Zhang, C. Xia, and L. Sun, “Graph-Bert: Only Attention is Needed for Learning Graph Representations,” *arXiv preprint*, 2020.
- [152] J. Zhang, Y. Dong, Y. Wang, J. Tang, and M. Ding, “ProNE: Fast and Scalable Network Representation Learning,” in *IJCAI*, 2019, pp. 4278–4284.
- [153] M. Zhang and Y. Chen, “Link Prediction Based on Graph Neural Networks,” in *NeurIPS*, 2018.
- [154] P. Zhang, Y. Yan, X. Zhang, L. Kang, C. Li, F. Huang, S. Wang, and S. Kim, “GPT4Rec: Graph prompt tuning for streaming recommendation,” in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*. Association for Computing Machinery, 2024, pp. 1774–1784. [Online]. Available: <https://doi.org/10.1145/3626772.3657720>
- [155] T. Zhang, F. Ladhak, E. Durmus, P. Liang, K. McKeown, and T. B. Hashimoto, “Benchmarking Large Language Models for News Summarization,” *Transactions of the Association for Computational Linguistics*, 2024.
- [156] W. Zhang, Y. Zhu, M. Chen, Y. Geng, Y. Huang, Y. Xu, W. Song, and H. Chen, “Structure Pretraining and Prompt Tuning for Knowledge Graph Transfer,” in *The Web Conference*, 2023, pp. 2581–2590.
- [157] Z. Zhang, Q. Liu, H. Wang, C. Lu, and C. Lee, “ProtGNN: Towards Self-Explaining Graph Neural Networks,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2022.
- [158] Z. Zhang, M. Liu, A. Wang, H. Chen, Z. Li, J. Bu, and B. He, “Collaborate to adapt: Source-free graph domain adaptation via bi-directional adaptation,” in *Proceedings of the ACM Web Conference 2024*, 2024, pp. 664–675. [Online]. Available: <https://doi.org/10.1145/3589334.3645507>
- [159] Z. Zhang, H. Li, Z. Zhang, Y. Qin, X. Wang, and W. Zhu, “Large graph models: A perspective,” *arXiv preprint*, 2023.
- [160] H. Zhao, A. Chen, X. Sun, H. Cheng, and J. Li, “All in one and one for all: A simple yet effective method towards cross-domain graph pretraining,” in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024.
- [161] H. Zhao, S. Liu, C. Ma, H. Xu, J. Fu, Z.-H. Deng, L. Kong, and Q. Liu, “GIMLET: A Unified Graph-Text Model for Instruction-Based Molecule Zero-Shot Learning,” in *Advances in Neural Information Processing Systems* 36, 2023.
- [162] H. Zhao, B. Yang, Y. Cen, J. Ren, C. Zhang, Y. Dong, E. Kharlamov, S. Zhao, and J. Tang, “Pre-Training and Prompting for Few-Shot Node Classification on Text-Attributed Graphs,” in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM, 2024, pp. 4467–4478. [Online]. Available: <https://doi.org/10.1145/3637528.3671952>
- [163] W. Zhao, Q. Wu, C. Yang, and J. Yan, “GraphGLOW: Universal and Generalizable Structure Learning for Graph Neural Networks,” in *KDD*, 2023, pp. 3525–3536.
- [164] Y. Zhao, B. Jiang, X. Wang, Q. Xu, and J. Tang, “Hegraphadapter: Tuning multi-modal vision-language models with heterogeneous graph adapter,” *arXiv preprint*, 2024. [Online]. Available: <https://arxiv.org/abs/2410.07854>
- [165] J. Zhu, Z. Ding, J. Yu, J. Tan, X. Li, and W. Qian, “RELIEF: Reinforcement Learning Empowered Graph Feature Prompt Tuning,” in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.1*. ACM, 2025, pp. 2159–2170. [Online]. Available: <https://doi.org/10.1145/3690624>

3709252

- [166] X. Zhu, H. Xue, Z. Zhao, W. Xu, J. Huang, M. Guo, Q. Wang, K. Zhou, I. Razzak, and Y. Zhang, “Llm as gnn: Graph vocabulary learning for text-attributed graph foundation models,” arXiv preprint, 2025. [Online]. Available: <https://arxiv.org/abs/2503.03313>
- [167] Y. Zhu, Y. Xu, F. Yu, Q. Liu, S. Wu, and L. Wang, “Graph Contrastive Learning with Adaptive Augmentation,” in The Web Conference, 2021, pp. 2069–2080.
- [168] Y. Zhu, H. Zhang, J. Yao, and H. Cao, “FFCL-KGC: Feature fusion contrastive learning via dual prompting for knowledge graph completion,” Expert Systems with Applications, vol. 299, p. 130117, 2026. [Online]. Available: <https://doi.org/10.1016/j.eswa.2025.130117>
- [169] Y. Zhu, J. Guo, and S. Tang, “SGL-PT: A Strong Graph Learner with Graph Prompt Tuning,” arXiv preprint, 2023.
- [170] Y. Zhu, Y. Wang, H. Shi, Z. Zhang, D. Jiao, and S. Tang, “GraphControl: Adding Conditional Control to Universal Graph Pre-trained Models for Graph Domain Transfer Learning,” in Proceedings of the ACM Web Conference 2024, 2024, pp. 539–550.
- [171] C. Zi, H. Zhao, X. Sun, Y. Lin, H. Cheng, and J. Li, “Prog: A Graph Prompt Learning Benchmark,” arXiv preprint, 2024. [Online]. Available: <https://arxiv.org/abs/2406.05346v2>
- [172] A. Zolnai-Lucas, J. Boylan, C. Hokamp, and P. Ghaffari, “STAGE: Simplified text-attributed graph embeddings using pre-trained LLMs,” in Proceedings of the 1st Workshop on Knowledge Graphs and Large Language Models (KaLLM 2024), 2024, pp. 92–104. [Online]. Available: <https://aclanthology.org/2024.kallm-1.10/>